WINTRIX ACADEMY PRESENTS

DCET Previous Year Question Papers & Solutions

2023 – 2026

COMPLETE 4-YEAR COLLECTION — DIPLOMA COMMON ENTRANCE TEST

GUIDANCE • STRATEGY • SUCCESS

400 Questions • 4 Years • Complete Answer Keys • Subject-Wise Organisation

FIRST EDITION — 2026

DCET Previous Year Question Papers & Solutions — 2023 to 2026

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Disclaimer: This book contains previous year question papers from the DCET examination conducted by the Karnataka Examination Authority (KEA). The solutions and explanations provided are prepared by Wintrix Academy for educational purposes. While every effort has been made to ensure accuracy, readers are advised to verify critical information independently.

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About Wintrix Academy

Wintrix Academy is a premier DCET mentorship platform founded with a singular mission: to empower Diploma students across Karnataka with the guidance, strategy, and resources they need to secure top ranks in the Diploma Common Entrance Test and gain admission to the state's most prestigious engineering institutions.

Built on the foundation of real experience and proven results, Wintrix Academy bridges the gap between aspiration and achievement through structured mentorship, comprehensive study materials, and a supportive learning ecosystem.

Mohammed Adnan

FOUNDER & CHIEF MENTOR

Mohammed Adnan completed his SSLC at CKS with 91.52% and pursued a Diploma in Electrical & Electronics Engineering, graduating with a CGPA of 9.8. He secured All-Karnataka DCET Rank 48 and 2nd Rank in Hassan District, earning admission to R.V. College of Engineering (RVCE) for Computer Science Engineering, where he currently maintains a CGPA of 9.19. His dedication to excellence has led to a placement at Red Hat, a renowned US-based technology company.

48	9.8	9.19	50+
DCET RANK	DIPLOMA CGPA	RVCE CGPA	STUDENTS

Jnanesh Gowda

DIRECTOR & BRAND ARCHITECT

Turning confusion into clarity and effort into results — that's what drives Wintrix forward. Jnanesh brings new-age learning strategies, purpose-driven outreach, structured content design, and results-engineered systems to ensure every student's success is built by design, not left to chance.

Our Promise: At Wintrix Academy, success isn't left to chance — it's engineered from day one. We provide live classes, recorded lectures, mock tests, rankings, 24/7 support, and a thriving community of DCET aspirants.

Preface

Dear DCET Aspirant,

Preparing for the Diploma Common Entrance Test is a journey that demands clarity, consistency, and the right resources. Having walked this very path and secured All-Karnataka Rank 48, I understand the challenges, the uncertainties, and the relentless effort it takes to achieve a top rank.

This comprehensive book has been meticulously crafted to serve as your ultimate companion in that journey. It contains the complete DCET Question Papers from 2023 to 2026 — four consecutive years — along with detailed, step-by-step solutions for all 400 questions. Every solution has been prepared with the goal of not just giving you the right answer, but helping you understand the reasoning behind it.

The questions for each year are organised subject-wise across five chapters — Engineering Mathematics, Statistics & Analytics, IT Skills, Fundamentals of Electrical & Electronics Engineering, and Project Management Skills — each carrying 20 marks. This structure mirrors the actual exam pattern and allows you to focus your preparation systematically.

With four years of papers at your fingertips, you can identify recurring patterns, frequently tested topics, and the evolving difficulty level of the exam. This kind of insight is invaluable for strategic preparation and time management.

I encourage you to attempt each question on your own before looking at the solution. Mark the questions you find challenging and revisit them until you've mastered the concept. Remember, every question you solve brings you one step closer to your dream college.

Wishing you all the very best for your DCET journey. May your hard work translate into the rank you deserve.

Mohammed Adnan

Founder, Wintrix Academy

DCET Rank 48 | RVCE CSE

Table of Contents

PRELIMINARIES

400 QUESTIONS • 4 YEARS • 5 SUBJECTS PER YEAR • 100 MARKS EACH

Introduction to D-CET

DCET — Diploma Common Entrance Test — is conducted by the Karnataka Examination Authority (KEA). The entrance exam is a requirement for enrolment in the second-year undergraduate engineering programmes offered through the lateral entry programme. In Karnataka, these programmes are offered at both day and evening colleges.

D-CET Exam Pattern

Duration	3 Hours
Maximum Marks	100
Type of Questions	Multiple Choice (MCQ)
Total Questions	100
Subjects Covered	5 Subjects

D-CET Tentative Important Dates

Application form starts	2nd Week of April
Last date for application submission	4th Week of April
D-CET exam date	Month of May
Counselling date	Month of June

Subject-wise Mark Distribution

1. Engineering Mathematics	20 Marks
2. Statistics and Analytics	20 Marks
3. IT Skills	20 Marks
4. Fundamentals of Electrical & Electronics Engineering	20 Marks
5. Project Management Skills	20 Marks
Total	**100 Marks**

2023

PART ONE

DCET 2023

100 Questions • 5 Subjects • 100 Marks

Previous Year Question Paper & Complete Solutions

01

CHAPTER ONE

Engineering Mathematics

20 Marks

Questions 1 – 20

Q.01

MATRICES

In a square matrix, if the elements above the principal diagonal are zero, then it is called

a) Identity matrix

b) Lower triangular matrix

c) Upper triangular matrix

d) Diagonal matrix

✓ ANSWER: (B)

In a square matrix, if all the elements above the principal diagonal are zero, it is called a **Lower Triangular Matrix**.

Q.02

DETERMINANTS

The value of x if the matrix $\begin{bmatrix} x-1 & 2 \\ 2 & 4 \end{bmatrix}$ is singular, is

a) 3

b) 8

c) −2

d) 2

✓ ANSWER: (D)

Given matrix $A = \begin{bmatrix} x-1 & 2 \\ 2 & 4 \end{bmatrix}$ is singular $\Longrightarrow |A| = 0$

$$(x-1)\cdot 4 - 2\cdot 2 = 0$$
$$4x - 4 - 4 = 0$$
$$4x = 8$$
$$\therefore x = 2$$

Q.03

MATRIX INVERSE

The inverse of the matrix $A = \begin{bmatrix} -1 & 0 \\ 5 & 7 \end{bmatrix}$ is

a) $-1/7 \begin{bmatrix} 7 & 0 \\ -5 & -1 \end{bmatrix}$

b) $1/7 \begin{bmatrix} 7 & 0 \\ 5 & 1 \end{bmatrix}$

c) $-1/7 \begin{bmatrix} -7 & 0 \\ 5 & -1 \end{bmatrix}$

d) $1/7 \begin{bmatrix} 7 & 0 \\ 5 & 1 \end{bmatrix}$

✓ ANSWER: (A)

Given $A = \begin{bmatrix} -1 & 0 \\ 5 & 7 \end{bmatrix}$

$$|A| = (-1)(7) - (0)(5) = -7$$

$$A^{-1} = (1/|A|) \cdot adj(A) = -1/7 \cdot \begin{bmatrix} 7 & 0 \\ -5 & -1 \end{bmatrix}$$

Q.04

EIGENVALUES

The eigenvalue of the matrix $A = \begin{bmatrix} 3 & 0 \\ 1 & 3 \end{bmatrix}$ is

a) 2, 2

b) −3, −3

c) 3, 3

d) −3, 3

✓ ANSWER: (C)

The characteristic equation of A is $|A - \lambda I| = 0$

$$\begin{vmatrix} 3-\lambda & 0 \\ 1 & 3-\lambda \end{vmatrix} = 0$$

$$(3-\lambda)(3-\lambda) - 0 = 0$$

$$(3-\lambda)^2 = 0$$

$$\therefore \lambda = 3, \lambda = 3$$

Q.05

STRAIGHT LINES

The two lines ax + by = c and a'x + b'y = c' are perpendicular if

a) ab' = ba'

b) aa' + bb' = 0

c) ab + a'b = 0

d) ab + ba = 0

✓ ANSWER: (B)

For line ax + by = c, slope $m_1 = -a/b$. For line a'x + b'y = c', slope $m_2 = -a'/b'$.

```
Lines are perpendicular when m₁ × m₂ = −1
(−a/b)(−a'/b') = −1
aa'/bb' = −1
∴ aa' + bb' = 0
```

Q.06

STRAIGHT LINES

The y-intercept of any line passing through the origin is

a) 0

b) 1

c) −1

d) 2

✓ ANSWER: (A)

The y-intercept of any line passing through the origin is zero, since the line crosses the y-axis at the point (0, 0).

Q.07

STRAIGHT LINES

Slope intercept form of straight line is

a) y + mx − c = 0

b) x = my + c

c) y = x + m

d) y = mx + c

✓ ANSWER: (D)

The slope-intercept form of a straight line is `y = mx + c`, where m is the slope and c is the y-intercept.

Q.08

STRAIGHT LINES

The tangent of the angle of intersection between two lines with slopes m_1 and m_2 is

a) $\left|\dfrac{m_1 + m_2}{1 - m_1m_2}\right|$

b) $\left|\dfrac{m_1 + m_2}{m_1 - m_2}\right|$

c) $\left|\dfrac{m_1 - m_2}{1 - m_1m_2}\right|$

d) $\left|\dfrac{m_1 - m_2}{1 + m_1m_2}\right|$

✓ ANSWER: (D)

The tangent of the angle of intersection between two lines with slopes m_1 and m_2 is

$$\tan\theta = \left|\dfrac{m_1 - m_2}{1 + m_1m_2}\right|$$

Q.09

TRIGONOMETRY

If the ladder is inclined to the wall at an angle of 135°, then the inclination in radians is

a) $\pi/4$

b) $\pi/2$

c) $3\pi/4$

d) $2\pi/3$

✓ ANSWER: (C)

Given angle $\theta = 135°$

$$\text{Angle in radians} = (\pi/180) \times 135 = 135\pi/180 = 3\pi/4 \text{ radians}$$

Q.10

TRIGONOMETRY

The value of $\sin 60° \cos 30° - \cos 60° \sin 30°$ is

a) 0

b) 1/2

c) $\sqrt{3}/2$

d) 1

✓ ANSWER: (B)

Using the formula $\sin(A - B) = \sin A \cos B - \cos A \sin B$:

$$\sin 60° \cos 30° - \cos 60° \sin 30° = \sin(60° - 30°) = \sin 30° = 1/2$$

Q.11

TRIGONOMETRY

The simplified value of (sin 3A + sin A) / (sin 3A − sin A) is

a) cot A tan 5A

b) tan A cot 2A

c) tan 2A cot A

d) tan 3A cot 2A

✓ ANSWER: (C)

Using transformation formulae:

```
sin C + sin D = 2 sin((C+D)/2) cos((C−D)/2)
sin C − sin D = 2 cos((C+D)/2) sin((C−D)/2)

(sin 3A + sin A) / (sin 3A − sin A)
= [2 sin 2A cos A] / [2 cos 2A sin A]
= tan 2A · cot A
```

Q.12

DIFFERENTIAL CALCULUS

If y = log x + sec 2x, then dy/dx is

a) 1/x + sec 2x tan x

b) 1/x + 2 sec 2x tan 2x

c) $1/x^2$ + sec x tan x

d) $1/x^2$ + sec 3x tan 2x

✓ ANSWER: (B)

```
y = log x + sec 2x
dy/dx = 1/x + 2 sec 2x · tan 2x
```

Q.13

DIFFERENTIAL CALCULUS

The derivative of the function (1 + x) / (1 − x) is

a) $1/(1-x)^2$

b) $-2/(1-x)^2$

c) $x/(1-x)^2$

d) $2/(1-x)^2$

✓ ANSWER: (D)

Using quotient rule: y = (ax+b)/(cx+d), then dy/dx = $(ad-bc)/(cx+d)^2$

```
y = (1+x)/(1−x)
dy/dx = [(1)(−1) − (1)(1)] · (−1) / (1−x)²
= 2/(1−x)²
```

Q.14

DIFFERENTIAL CALCULUS

Find the second order derivative of $y = e^{2x} - e^{x}$

a) $4e^{2x} - e^{x}$

b) $4e^{2x} + e^{x}$

c) $-4e^{2x} - e^{x}$

d) $4e^{x} + e^{x}$

✓ ANSWER: (A)

$$y = e^{2x} - e^{x}$$
$$dy/dx = 2e^{2x} - e^{x}$$
$$d^2y/dx^2 = 4e^{2x} - e^{x}$$

Q.15

APPLICATIONS

The equation of the tangent to the curve $y = 2x^2 + x$ at (1, 2) is

a) $5x - y - 3 = 0$

b) $5x + y + 3 = 0$

c) $5x + y - 6 = 0$

d) $5x + y + 6 = 0$

✓ ANSWER: (A)

$$y = 2x^2 + x$$
$$dy/dx = 4x + 1$$
At (1, 2): $m = dy/dx|_{(1)} = 4(1) + 1 = 5$

Equation of tangent: $y - y_1 = m(x - x_1)$
$$y - 2 = 5(x - 1)$$
$$y - 2 = 5x - 5$$
$$\therefore 5x - y - 3 = 0$$

Q.16

INTEGRAL CALCULUS

The value of $\int(\sec x \tan x + \sec^2 x)\,dx$ is

a) $\sec 2x + \tan x + c$

b) $\sec x + \operatorname{cosec} x + c$

c) $\sec x + \tan x + c$

d) $\sec 2x - \operatorname{cosec}^2 x + c$

✓ ANSWER: (C)

$$\int(\sec x \tan x + \sec^2 x)\,dx = \sec x + \tan x + c$$

Q.17

INTEGRAL CALCULUS

The value of $\int(2x^3 + 3x^2 + 2x)^{10} (3x^2 + 3x + 1)\,dx$ is

a) $1/22\cdot(6x + 3x^2 + 2x)^{11} + c$

b) $1/22\cdot(2x^3 + 3x^2 + 2x)^{11} + c$

c) $1/12\cdot(2x^3 + 3x^2 - 2x)^{12} + c$

d) $1/12\cdot(2x^3 + 3x^2 - 2x)^{12} + c$

✓ ANSWER: (B)

Using the formula $\int[f(x)]^n f'(x)\,dx = [f(x)]^{n+1}/(n+1) + c$

```
Let f(x) = 2x³ + 3x² + 2x
f'(x) = 6x² + 6x + 2 = 2(3x² + 3x + 1)

∫(2x³+3x²+2x)¹⁰(3x²+3x+1)dx
= ½ ∫(2x³+3x²+2x)¹⁰(6x²+6x+2)dx
= ½ · (2x³+3x²+2x)¹¹/11 + c
= 1/22 · (2x³+3x²+2x)¹¹ + c
```

Q.18

DEFINITE INTEGRALS

The value of $\int_0^{\pi/4} \tan^2 x\,dx$ is

a) $1 + \pi/4$

b) $1 + 4/\pi$

c) $1 - \pi/4$

d) $1 - 4/\pi$

✓ ANSWER: (C)

```
∫₀^(π/4) tan²x dx = ∫₀^(π/4) (sec²x − 1) dx
= [tan x − x]₀^(π/4)
= (tan π/4 − π/4) − (tan 0 − 0)
= 1 − π/4
```

Q.19

APPLICATIONS

The area bounded by $y = \sin x$ and x-axis from $x = 0$ to $x = \pi$ is

a) 2

b) −2

c) 3

d) 1

✓ ANSWER: (A)

Area = $\int_0^\pi \sin x \, dx = [-\cos x]_0^\pi$

$= -\cos \pi - (-\cos 0) = -(-1) + 1 = 1 + 1 = 2$

Q.20

TRIGONOMETRY

The value of $\tan 45° \cdot \cot 225° + \tan^2 60°$ is

a) −4

b) 4

c) 2

d) 3

✓ ANSWER: (B)

$\tan 45° \cdot \cot 225° + \tan^2 60°$

$= 1 \times \cot(180° + 45°) + (\sqrt{3})^2$

$= 1 \times \cot 45° + 3$

$= (1 \times 1) + 3 = 1 + 3 = 4$

02

CHAPTER TWO

Statistics & Analytics

20 Marks

Questions 21 – 40

Q.21

DATA TYPES

__________ is an example of quantitative data.

a) Volume
b) Words
c) Symbols
d) Colour

✓ ANSWER: (A)

The data which can be numerically measurable is called **quantitative data**. "Volume" is measured in numbers, hence it is quantitative data.

Q.22

DATA CLEANING

Data cleaning is the process of

a) removing viruses
b) correctly formatting data
c) removing duplicate data
d) properly formatting data

Grace Marks

Data cleaning is the process of fixing or removing incorrect, corrupted, incorrectly formatted, duplicate or incomplete data within a dataset.

Q.23

DATA COLLECTION

__________ is not a data collection tool.

a) Word
b) Focus Group Discussion
c) Survey
d) Questionnaires

✓ ANSWER: (A)

Data collection tools include focus group discussion, survey method, questionnaires, observation, and interview method. **Word** is not a data collection tool.

Q.24

DATA VISUALIZATION

The graph of cumulative frequency is called

a) Frequency polygon

b) Histogram

c) Cumulative frequency polygon

d) Frequency histogram

Grace Marks

The cumulative frequencies can be plotted on a **cumulative frequency polygon** or cumulative frequency curve (Ogive curve).

Q.25

FREQUENCY

To calculate percentage frequency, we use __________ formula.

a) P.f. = (f × n) + 100

b) P.f. = (f ÷ n) × 100

c) P.f. = (100) ÷ (f × n)

d) P.f. = (100) × (f ÷ n)

✓ **ANSWER: (B) & (D)**

Percentage frequency = `(f/n) × 100` where f is the frequency and n is the total number of observations.

Q.26

CENTRAL TENDENCY

If X_1, X_2, X_3, ..., X_n are the observations of a given data, then the mean of the observations will be:

a) Total number of observations / Sum of observations

b) Sum of observations + Total number of observations

c) Sum of observations / Total number of observations

d) Total number of observations − Sum of observations

✓ **ANSWER: (C)**

Mean = `Sum of observations / Total number of observations`

Q.27

CLASS INTERVALS

The end points of a class interval are the __________ and __________ values that a variable can take.

a) Lowest and Highest

b) Minimum and Maximum

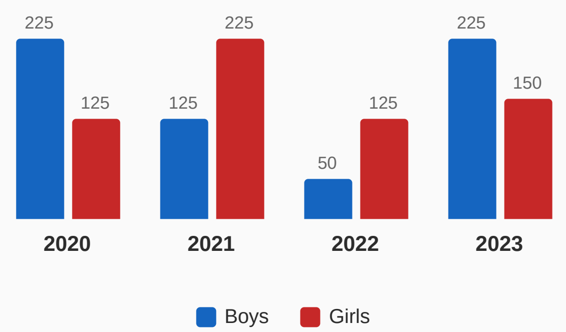
c) Numeral and Average

d) Mean and Mode

✓ ANSWER: (A)

If [10–20] is a class interval, the end points are 10 and 20, which are the **lowest and highest** values of the variable.

The graph given below, shows the participation of students of a school in outdoor games in different years. By using the data given below, answer the following questions (28 – 29):



Q.28

DATA INTERPRETATION

In which years did the girls participate more than the boys?

a) 2020, 2023

b) 2021, 2022

c) 2022, 2023

d) 2020, 2021

✓ ANSWER: (B)

From the data: In 2021 (Girls=225 > Boys=125) and 2022 (Girls=125 > Boys=50), girls participated more than boys.

Q.29

DATA INTERPRETATION

In which two years did an equal number of boys participate?

a) 2020, 2021

b) 2020, 2022

c) 2020, 2023

d) 2021, 2022

✓ ANSWER: (C)

In 2020 boys participation = 225, and in 2023 boys participation = 225. Hence equal.

Q.30

EXCEL

To find third quartile in Excel, we use ___________ formula.

a) =QUARTER (3, Range)

b) =QUARTILE (3, Range)

c) =QUARTER (Range, 3)

d) =QUARTILE (Range, 3)

✓ ANSWER: (D)

The formula for finding the 3rd quartile in Excel is `=QUARTILE(Range, 3)`

Q.31

PERCENTILE

The percentile divides a series into ___________ equal parts.

a) fifty

b) twenty

c) ten

d) hundred

✓ ANSWER: (D)

Data which divides into **100 equal parts** is called percentile. It is denoted by P.

Q.32

QUARTILES

The value of first quartile is 23 and interquartile range of set of data is 20. The value of third quartile is

a) 23

b) 33

c) 42

d) 43

✓ ANSWER: (D)

$Q_1 = 23$, $IQR = Q_3 - Q_1 = 20$

$Q_3 - 23 = 20$

$Q_3 = 20 + 23 = 43$

Q.33

MEAN

The algebraic sum of the deviations of a frequency distribution from its mean is always

a) a non-zero number

b) zero

c) less than zero

d) greater than zero

✓ ANSWER: (B)

The property of arithmetic mean states that the algebraic sum of the deviations from the mean is always equal to **zero**. i.e., $\Sigma(X - \bar{X}) = 0$

Q.34

EXCEL

The Excel formula for "Mean" is

a) =MEDIAN (array of numbers)

b) =AVERAGE (array of numbers)

c) =MEAN (array of numbers)

d) =MODE (array of numbers)

✓ ANSWER: (B)

The Excel formula for "mean" is `=AVERAGE(array of numbers)`

Q.35

PYTHON

What is output syntax in Python?

a) Print(" ")

b) PRINT(" ")

c) print(" ")

d) Printf (" ")

✓ ANSWER: (C)

In Python, `print(" ")` statement is used for output. Python is case-sensitive.

Q.36

PYTHON

"str" is a __________

a) Text Type

b) Numeric Type

c) Binary Type

d) Sequence Type

✓ ANSWER: (A)

`str` is a Text Type in Python.

Q.37

PYTHON

In Python, __________ standard data types are commonly used.

a) three

b) five

c) ten

d) four

Grace Marks

In Python, standard data types are: Numeric, Dictionary, Boolean, Set, and Sequence types.

Q.38

PYTHON

The result of Python program gets displayed in __________

a) IDLE Shell 3.9.1 window

b) IDLE Shell 3.1.9 window

c) ILDE Shell 3.9.1 window

d) IELD Shell 3.9.1 window

✓ ANSWER: (A)

IDLE 3.9.1 is the Integrated Development and Learning Environment for Python.

Q.39

PYTHON

Which one of the following quotations in Python does not accept the quotes to denote strings?

a) (' ')

b) (" ")

c) ( )

d) (""" """)

✓ ANSWER: (C)

In Python, it accepts single quotes, double quotes, and triple quotes. Parentheses `( )` without quotes do not denote strings.

Q.40

PYTHON

In Python, ___________ is used to end the physical line or ignore the comment.

a) **

b) #

c) &

d) \

✓ ANSWER: (B)

In Python, the `#` character is used to start a comment. The comment continues after # until the end of the line.

03

CHAPTER THREE

IT Skills

20 Marks

Questions 41 – 60

Q.41

CYBER SECURITY

Which of the following is not a cyber crime?

a) Cryptography

b) Denial of Service

c) Man-in-the-middle attack

d) Phishing

✓ ANSWER: (A)

Cryptography is the process of hiding or coding information so that only the intended recipient can read it. It is a security measure, not a cyber crime.

Q.42

CYBER SECURITY

DoS is abbreviated as

a) Denial of Service

b) Distribution of Server

c) Distribution of Service

d) Denial of Server

✓ ANSWER: (A)

DoS stands for **Denial-of-Service**.

Q.43

CYBER SECURITY

__________ protects interconnected systems including hardware, software and programs and data from cyber attacks.

a) Cyber Security

b) Computer Security

c) Resource Security

d) Hardware Security

✓ ANSWER: (A)

Cyber Security is the protection of interconnected systems including hardware, software, programs, and data from cyber attacks.

Q.44

NETWORKING

Basic functionality of the network device firewall is:

a) scans mobile applications

b) monitoring database

c) privatizes the computers

d) monitoring incoming and outgoing networks

✓ ANSWER: (D)

A **firewall** is a network security device that monitors incoming and outgoing network traffic and decides whether to allow or block specific traffic based on a defined set of security rules.

Q.45

PROGRAMMING

An algorithm represented in the form of programming language is called:

a) Flowchart

b) Pseudocode

c) Program

d) Instruction

✓ ANSWER: (C)

An algorithm represented in the form of a programming language is called a **program**.

Q.46

FLOWCHARTS

The ___________ symbol is used when the flowchart is starting or ending.

a) Connector/Arrow

b) Terminal box/Rounded rectangle

c) Input/Output

d) Process

✓ ANSWER: (B)

The **Terminator Symbol** (rounded rectangle) represents the start and end points of a flowchart.

Q.47

APP DEVELOPMENT

MIT App Inventor allows user to

a) Create web application

b) Build Android application

c) Create System Software

d) Develop Operating System

✓ ANSWER: (B)

MIT App Inventor allows users to build fully functional apps for smartphones and tablets.

Q.48

SCRATCH

What is the function of the "when green flag clicked" command block?

a) Points Sprite in the specified direction

b) If condition is true, runs the blocks inside

c) Runs the script

d) Stops the execution of script

✓ ANSWER: (C)

The "when green flag clicked" block triggers the execution of the script or program attached to it.

Q.49

HTML

The correct sequence of HTML tags for starting a webpage is

a) Head, Title, Html, Body

b) Html, Head, Title, Body

c) Html, Body, Title, Head

d) Html, Title, Head, Body

✓ ANSWER: (B)

The correct sequence of HTML tags to start a webpage is `<html>, <head>, <title>, <body>`

Q.50

WEB

Web server:

a) is a computer system that delivers web pages

b) is delivery news

c) provides options for those seeking real-time discussions

d) prints documents

✓ ANSWER: (A)

A **web server** stores and delivers the content for a website.

Q.51

WEB

Which of the following is used to style the appearance of web pages?

a) HTML

b) JavaScript

c) PHP

d) CSS

✓ ANSWER: (D)

CSS is used to define styles for web pages, including design, layout, and variations in display for different devices and screen sizes.

Q.52

WEB

Which of the following is an example of a web browser?

a) Google

b) Firefox

c) Apache

d) MySQL

✓ ANSWER: (B)

Firefox is an example of a web browser.

Q.53

WORKFLOW

Which of the following is an open source and free workflow management software?

a) Trello

b) MS Excel

c) Windows

d) Linux

✓ ANSWER: (A)

Trello is the visual tool that empowers teams to manage any type of project, workflow, or task tracking.

Q.54

ERP

ERP package will handle ___________ business functionality/functionalities.

a) One

b) Two

c) Three

d) Multiple/all

✓ ANSWER: (D)

ERP systems support all aspects of modules and provide transparency into the complete business process.

Q.55

ORGANIZATION

___________ is a visual diagram of a company that describes what employees do, whom they report to and how decisions are made across the business.

a) Physical Structure

b) Organizational Structure

c) Logical Structure

d) Hybrid Structure

✓ ANSWER: (B)

Organisational structure describes the entire structure of an organisation and work structure of employees.

Q.56

WORKFLOW

___________ is a methodology used in system analysis to identify, clarify, and organize system requirements.

a) Workflow

b) Use case

c) Algorithm

d) Software

✓ ANSWER: (A)

Workflow helps in identifying, clarifying, and organising requirements of a system.

Q.57

IOT

Which of the following is not an application of IoT?

a) Web browser

b) Smart home

c) Smart city

d) Self-driven cars

✓ ANSWER: (A)

Smart home, smart city, and self-driving cars are applications of IoT, but a **web browser** is not an IoT application.

Q.58

CLOUD

Which of the following is not a cloud service option?

a) VaaS

b) IaaS

c) PaaS

d) SaaS

ANSWER: (A)

PaaS, SaaS, and IaaS are the three cloud services. **VaaS** (Voice as a Service) is not a standard cloud service model.

Q.59

CLOUD

How many types of services are offered by cloud computing to the users?

a) 2

b) 4

c) 3

d) 5

ANSWER: (C)

Cloud computing offers **3 types** of services: PaaS, SaaS, and IaaS.

Q.60

CLOUD

Combination of Public and Private deployment is called

a) Hybrid

b) Hyper

c) Public

d) Private

ANSWER: (A)

Hybrid cloud refers to a mixed computing, storage, and services environment made up of on-premises infrastructure, private cloud services, and a public cloud.

04

CHAPTER FOUR

Fundamentals of Electrical & Electronics Engineering

20 Marks

Questions 61 – 80

Q.61

ELECTRICAL BASICS

Unit of electrical power is

a) Volt
b) Watt
c) Watt-hour
d) Ampere-hour

✓ ANSWER: (B)

A **Watt** is the unit of electrical power equal to one ampere under the pressure of one volt. 1 W is the power consumed by a device that carries 1 A of current when operated at a potential difference of 1 V.

Q.62

EARTHING

In pipe earthing, the diameter of GI pipe embedded in the pit is

a) 32 mm
b) 38 mm
c) 48 mm
d) 56 mm

✓ ANSWER: (B)

Pipe electrodes shall not be smaller than **38 mm** internal diameter if made of galvanized iron or steel, and 100 mm internal diameter if made of cast iron.

Q.63

CIRCUITS

If a resistor of 100 ohms is connected in series with a parallel combination of two 200 ohms resistors, then the effective resistance is

a) 200 ohms
b) 250 ohms
c) 350 ohms
d) 150 ohms

✓ ANSWER: (A)

Series resistor: $R_1 = 100\ \Omega$

Parallel: $R_2 = 200\ \Omega$, $R_3 = 200\ \Omega$

$R_p = (200 \times 200)/(200 + 200) = 40000/400 = 100\ \Omega$

Effective $= R_p + R_1 = 100 + 100 = 200\ \Omega$

Q.64

OHM'S LAW

If a resistor of 20 Ω is connected across a source of 5 volts DC supply, then the current in the circuit is

a) 1 Ampere

b) 4 Amperes

c) 0.5 Amperes

d) 0.25 Amperes

✓ ANSWER: (D)

```
R = 20 Ω, V = 5 V
I = V/R = 5/20 = 0.25 Amperes
```

Q.65

AC CIRCUITS

Power factor is

a) ratio of resistance to inductance

b) ratio of apparent power to true power

c) ratio of resistance to impedance

d) ratio of inductance to capacitance

✓ ANSWER: (C)

In AC circuits, the **power factor** is defined as the ratio of resistance (R) to impedance (Z).

`cos φ = R/Z`

Q.66

3-PHASE

The phase-to-neutral voltage in a 3-phase star connected system is 230 V. The phase-to-phase (line-to-line) voltage is

a) 230 V

b) 398.37 V

c) 400 V

d) 440 V

✓ ANSWER: (B)

```
V_L = √3 × V_ph = √3 × 230 = 398.37 V
```

Q.67

AC THEORY

The time period of an AC wave at frequency of 50 Hz is

- a) 2 milliseconds
- b) 10 milliseconds
- c) 20 milliseconds
- d) 50 milliseconds

✓ **ANSWER: (C)**

```
T = 1/f = 1/50 = 0.02 s = 20 milliseconds
```

Q.68

PROTECTION

The type of fuse used for domestic purpose is

- a) HRC fuse
- b) Kitkat or rewirable fuse
- c) Ceramic cartridge fuse
- d) Glass cartridge fuse

✓ **ANSWER: (B)**

The **Kitkat or rewirable fuse** is used for domestic purposes. It safeguards the circuit and appliances from damage due to voltage fluctuations and short circuits.

Q.69

PROTECTION

MCCB stands for

- a) Molded Case Circuit Breaker
- b) Miniature Case Circuit Breaker
- c) Maximum Current Circuit Breaker
- d) Minimum Current Circuit Breaker

✓ **ANSWER: (A)**

MCCB stands for **Molded Case Circuit Breaker**.

Q.70

PROTECTION

ELCB is used for detecting current leakage

- a) above 8 kVA
- b) below 5 kVA
- c) above 5 kVA
- d) below 8 kVA

Grace Marks

ELCB (Earth Leakage Circuit Breaker) is used for detecting current leakage to earth.

Q.71

MACHINES

A static machine which transfers electrical power from one circuit to another circuit without changing the frequency is called

a) DC machine

b) Alternator

c) Induction motor

d) Transformer

✓ ANSWER: (D)

A **Transformer** is designed to either increase or decrease AC voltage between circuits while maintaining the same frequency.

Q.72

MOTORS

The initial type of connection of the motor windings when started with a star delta starter is

a) star connection

b) delta connection

c) series

d) parallel

✓ ANSWER: (A)

During starting, the stator winding is **star connected**, which provides less starting torque and puts less mechanical and electrical stress on the system.

Q.73

MOTORS

The cause for a 3-phase motor producing inadvertent mechanical noise is:

a) Interchanged supply terminals

b) High load on motor

c) High voltage on motor winding

d) Incorrect coupling

✓ ANSWER: (D)

If the coupling is not aligned perfectly, it can cause excessive wear and eventually lead to failure. **Incorrect coupling** is a primary cause of mechanical noise.

Q.74

BATTERIES

Cell is an __________ device

a) electro-mechanical

b) electro-chemical

c) electro-magnetic

d) electro-dynamic

✓ ANSWER: (B)

An **electrochemical** cell generates electrical energy from chemical reactions or uses electrical energy to facilitate chemical reactions.

Q.75

BATTERIES

The most commonly used battery in electric vehicles is

a) Lithium-ion battery

b) Lead-acid battery

c) Nickel-Cadmium battery

d) Alkaline rechargeable battery

✓ ANSWER: (A)

Lithium-ion batteries are popular due to their light weight, high energy density, low self-discharge, no memory effect, and ability to recharge.

Q.76

DIGITAL ELECTRONICS

Digital signals are characterized by

a) Continuous voltage levels

b) Infinite resolution

c) Discrete voltage levels

d) Variable voltage levels

✓ ANSWER: (C)

A digital signal can only take on one value from a finite set of possible values at a given time — **discrete voltage levels**.

Q.77

BOOLEAN ALGEBRA

According to rules and laws of Boolean Algebra, A + A =

a) 2A

b) A

c) 1

d) A^2

✓ ANSWER: (B)

By the **Idempotent Law** of Boolean Algebra: A + A = A and A · A = A

Q.78

ELECTRONICS

Photo diode is used in which of the following applications?

a) Voltage regulation

b) Temperature measurement

c) Light detection

d) Radio Frequency (RF) Amplification

ANSWER: (C)

Photodiodes generate current flow directly depending on light intensity, making them suitable for **light detection** and optical signal detection.

Q.79

RESISTORS

If a resistor is having a colour band as first band = Brown, second band = Black and third band = Red, then the resistance value is

a) 1 kΩ

b) 10 kΩ

c) 100 Ω

d) 100 kΩ

ANSWER: (A)

```
1st band (Brown) = 1
2nd band (Black) = 0
3rd band (Red) = multiplier × 100
Resistance = 10 × 100 = 1000 Ω = 1 kΩ
```

Q.80

The binary equivalent of the decimal number 9 is

a) 1001

b) 1000

c) 1100

d) 1010

✓ ANSWER: (A)

$9 \div 2 = 4$ remainder 1
$4 \div 2 = 2$ remainder 0
$2 \div 2 = 1$ remainder 0
$1 \div 2 = 0$ remainder 1
$\therefore (9)_{10} = (1001)_2$

05

CHAPTER FIVE

Project Management Skills

20 Marks

Questions 81 – 100

Q.81

PM BASICS

Project Management is a combination of

a) human and non-human resources
b) only human resources
c) only non-human resources
d) no resources at all

✓ ANSWER: (A)

Project management involves the coordination and utilization of both **human resources** (managers, team members, stakeholders) and **non-human resources** (tools, equipment, technology, materials, finances).

Q.82

CONSULTANTS

The consultant who is appointed to carry out the project work is

a) Compound house consultant
b) In-house consultant
c) Out-house consultant
d) Bridge consultant

✓ ANSWER: (C)

An **outsourced (out-house) consultant** is someone hired from outside the organization to provide expertise, guidance, and support for a specific project.

Q.83

PROJECT TYPES

The type of project which requires minimum amount of capital is

a) Crash project
b) Normal project
c) Disaster project
d) Consultant project

✓ ANSWER: (B)

Normal projects are allowed to take their normal time, with minimum capital and no sacrifice in quality.

Q.84

PROJECT TYPES

Projects like building a hospital, a park, a playground and government projects like highway construction are examples of

a) Social needs

b) Customer needs

c) Market needs

d) Ecological needs

✓ ANSWER: (A)

Hospitals, parks, schools, and road construction are basic requirements for society — hence they are **social need** projects.

Q.85

ABBREVIATIONS

In Project Management, WBS stands for

a) Work Breakdown Structure

b) Waste Breakdown Structure

c) Window Breakdown Structure

d) Wireless Breakdown Structure

✓ ANSWER: (A)

WBS stands for **Work Breakdown Structure**.

Q.86

PEP

The first step in Project Execution Plan is

a) Work packaging plan

b) Contracting plan

c) Organization plan

d) Procedure plan

✓ ANSWER: (B)

Under the four sub-plans of PEP, the **contracting plan** is the first step. It is necessary to contract out areas where the owner company does not have inherent competence.

Q.87

TEAMS

The team which gives the idea to start a project is

a) Core project team

b) Full project team

c) Advising project team

d) Initial project team

✓ ANSWER: (D)

The **initial project team** consists of specific people who initially conceive the idea of starting a project.

Q.88

ABBREVIATIONS

In PEP, the letter 'E' stands for

a) Execution

b) Estimation

c) Evaluation

d) Enthusiasm

✓ ANSWER: (A)

PEP stands for Project **Execution** Plan.

Q.89

PROJECT LIFE CYCLE

In Project Life Cycle, more time is required for

a) Project closure

b) Project initiation

c) Project execution

d) Project planning

✓ ANSWER: (C)

The **execution phase** consumes maximum time to complete work in a project life cycle.

Q.90

INNOVATION

Innovation is the hallmark of every project. Here innovation means

a) New ideas

b) Project success

c) Professional approach

d) Project Management

✓ ANSWER: (A)

Innovation refers to the process of introducing **new ideas**, methods, products, services, or approaches that create value and improve upon existing practices.

Q.91

PROJECT LIFE CYCLE

The Project Life Cycle Curve indicates

a) Work packaging

b) Number of workers in the project

c) Growth, maturity and decline

d) Project manual

✓ ANSWER: (C)

The project life cycle curve explains how a project is initiated, its **growth, maturity, and decline** (completion).

Q.92

QUALITY

In Project Management, taking actions to measure the quality accurately is the function of

a) Quality management

b) Cost management

c) Review management

d) Risk management

✓ ANSWER: (A)

Quality Management is the method to measure quality accurately in Project Management.

Q.93

PLANNING

Project Planning Methodology involve

a) Planning by obligation

b) Planning by incentive and direction

c) Planned initiation

d) Changing the project policies

✓ ANSWER: (B)

Project planning involves creating incentives to motivate team members and providing clear guidance and **direction** to the project team.

Q.94

OBJECTIVES

Identify the incorrect statement below:

a) Project objective should be specific

b) Project objective should be realistic

c) Project objective should not be framed timely

d) Project objective should be measurable

✓ ANSWER: (C)

Project objectives must be specific, measurable, and realistic. They should be framed within a timeline — so the statement "should not be framed timely" is **incorrect**.

Q.95

TOOLS

WBS, PEP and PPM are the tools used to design

a) Project plan

b) Project work system

c) Project diary

d) Project direction

✓ ANSWER: (B)

Project Work System classifies into Work Breakdown Structure, Project Execution Plan, and Project Procedure Manual.

Q.96

PLANNING TOOLS

The earliest method used for planning of project was

a) CPM

b) PERT

c) Bar Chart

d) Milestone Chart

✓ ANSWER: (C)

The earliest method used for project planning is the **Bar Chart** (also known as Gantt Chart).

Q.97 ABBREVIATIONS

The expansion of PERT is

a) Programme Evaluation and Review Technique

b) Project Estimation and Review Technique

c) Project Estimation and Resource Technology

d) Performance Estimation and Resource Tool

ANSWER: (A)

PERT stands for **Programme Evaluation and Review Technique**.

Q.98 PLANNING TOOLS

For non-repetitive projects __________ tool is used in production planning and scheduling.

a) CPM

b) PERT

c) Both CPM and PERT

d) Bar Chart

ANSWER: (B)

For non-repetitive projects, **PERT** is the tool used in project planning and scheduling.

Q.99 REVIEW

The purpose of conducting a project review is

a) To finance the project

b) To initiate the project

c) To develop the project scope

d) To assess project performance

ANSWER: (D)

Project review helps in evaluating and **assessing project performance**.

Q.100

REVIEW

A project review does not contain

a) Performance evaluation

b) Evaluating the capital budget

c) Data collection

d) Initial review

✓ ANSWER: (C)

Data collection is not a step of project review.

2024

PART TWO

DCET 2024

100 Questions • 5 Subjects • 100 Marks

Previous Year Question Paper & Complete Solutions

01

CHAPTER ONE

Engineering Mathematics

20 Marks

Questions 1 – 20

Q.01

MATRICES

If $A = \begin{bmatrix} 3 & -2 \\ 4 & 1 \end{bmatrix}$ and $A - B = \begin{bmatrix} 5 & -3 \\ 4 & 6 \end{bmatrix}$, then matrix B is

a) $\begin{bmatrix} -3 & 2 \\ -4 & -1 \end{bmatrix}$

b) $\begin{bmatrix} -5 & 3 \\ -4 & -6 \end{bmatrix}$

c) $\begin{bmatrix} -2 & 1 \\ 0 & -5 \end{bmatrix}$

d) $\begin{bmatrix} 2 & -1 \\ 0 & 5 \end{bmatrix}$

✓ **ANSWER: (C)**

$$B = A - (A - B) = \begin{bmatrix} 3 & -2 \\ 4 & 1 \end{bmatrix} - \begin{bmatrix} 5 & -3 \\ 4 & 6 \end{bmatrix}$$

$$B = \begin{bmatrix} 3-5 & -2+3 \\ 4-4 & 1-6 \end{bmatrix} = \begin{bmatrix} -2 & 1 \\ 0 & -5 \end{bmatrix}$$

Q.02

CRAMER'S RULE

If $\begin{bmatrix} 3 & -4 \\ 9 & 2 \end{bmatrix} \begin{bmatrix} x & y \end{bmatrix} = \begin{bmatrix} 10 & 2 \end{bmatrix}$, then the values of x and y are

a) $\left(\frac{2}{3}, -2\right)$

b) $\left(2, \frac{2}{3}\right)$

c) $\left(-2, \frac{2}{3}\right)$

d) $\left(-\frac{2}{3}, 2\right)$

✓ **ANSWER: (A)**

$3x - 4y = 10 \;...\; (i)$

$9x + 2y = 2 \;...\; (ii)$

Multiply (i) by 2: $6x - 8y = 20$

Multiply (ii) by 4: $36x + 8y = 8$

Adding: $42x = 28 \rightarrow x = \frac{2}{3}$

Substituting: $3(2/3) - 4y = 10 \rightarrow y = -2$

Q.03

MATRIX INVERSE

If $A = \begin{bmatrix} -2 & 6 \\ -2 & 3 \end{bmatrix}$, then A^{-1} is

a) $\frac{1}{18}\begin{bmatrix} 3 & -6 \\ 2 & -2 \end{bmatrix}$

b) $\frac{1}{6}\begin{bmatrix} 3 & -6 \\ 2 & -2 \end{bmatrix}$

c) $\begin{bmatrix} 3 & -6 \\ 2 & -2 \end{bmatrix}$

d) $\frac{1}{6}\begin{bmatrix} -2 & 6 \\ -2 & 3 \end{bmatrix}$

✓ ANSWER: (B)

$$\begin{vmatrix} -2 & 6 \\ -2 & 3 \end{vmatrix} = (-2)(3) - (6)(-2) = -6 + 12 = 6$$

$$\text{adj}(A) = \begin{bmatrix} 3 & -6 \\ 2 & -2 \end{bmatrix}$$

$$A^{-1} = \frac{1}{|A|} \cdot \text{adj}(A) = \frac{1}{6}\begin{bmatrix} 3 & -6 \\ 2 & -2 \end{bmatrix}$$

Q.04

LINEAR EQUATIONS

If x + y = 3 and 2x + 3y = 8 are the linear equations, then x and y are

a) (1, −2)

b) (1, 2)

c) (−1, −2)

d) (−1, 2)

✓ ANSWER: (B)

x + y = 3 → multiply by 2: 2x + 2y = 6

2x + 3y = 8

Subtracting: -y = -2 → y = 2

x = 3 - 2 = 1 ∴ (x, y) = (1, 2)

Q.05

STRAIGHT LINES

If the straight line is $2y = 3x + 4$, then the y-intercept is

a) 4

b) 2

c) $-\frac{4}{3}$

d) -2

✓ ANSWER: (B)

$2y = 3x + 4 \to 3x - 2y + 4 = 0$

$\text{y-intercept} = \frac{-c}{b} = \frac{-4}{-2} = 2$

Q.06

STRAIGHT LINES

The acute angle of inclination made by the line $3x - 3y - 2 = 0$ with positive x-axis is

a) 135°

b) 30°

c) 45°

d) 0°

✓ ANSWER: (C)

$\text{Slope } m = \frac{-a}{b} = \frac{-3}{-3} = 1$

$\tan\theta = 1 \to \theta = 45°$

Q.07

STRAIGHT LINES

The slope of a straight line passing through the points (1, 2) and (5, −3) is

a) $\frac{5}{4}$

b) $\frac{1}{6}$

c) $-\frac{1}{4}$

d) $-\frac{5}{4}$

✓ ANSWER: (D)

$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{-3 - 2}{5 - 1} = -\frac{5}{4}$

Q.08

STRAIGHT LINES

If the slope of the straight line is $-\frac{3}{2}$ and y-intercept is $\frac{5}{2}$, then the equation of the straight line is

a) $3x + 2y - 5 = 0$

b) $3x - 2y - 5 = 0$

c) $3x + 2y + 5 = 0$

d) $3x - 2y + 5 = 0$

✓ ANSWER: (A)

$$y = mx + c = \left(-\frac{3}{2}\right)x + \frac{5}{2}$$

$$2y = -3x + 5 \rightarrow 3x + 2y - 5 = 0$$

Q.09

STRAIGHT LINES

The line perpendicular to the straight line $5x - y + 1 = 0$ and passing through the point (2, 3) is

a) $x - 5y - 17 = 0$

b) $x + 5y - 13 = 0$

c) $x + 5y - 17 = 0$

d) $x + 5y + 13 = 0$

✓ ANSWER: (C)

Slope of $5x - y + 1 = 0$ is $m = 5$

Slope of perpendicular $= -\frac{1}{5}$

$$y - 3 = \left(-\frac{1}{5}\right)(x - 2)$$

$$5y - 15 = -x + 2 \rightarrow x + 5y - 17 = 0$$

Q.10

TRIGONOMETRY

The value of cot 660° is

a) $-\frac{1}{\sqrt{3}}$

b) $\sqrt{3}$

c) $\frac{1}{\sqrt{3}}$

d) $-\sqrt{3}$

✓ ANSWER: (A)

$\cot 660° = \cot(360° + 300°) = \cot 300°$

$= \cot(270° + 30°) = -\tan 30° = -\frac{1}{\sqrt{3}}$

Q.11

TRIGONOMETRY

The value of cos 75° is

a) $\frac{\sqrt{3}+1}{2\sqrt{2}}$

b) $\frac{\sqrt{3}+1}{\sqrt{3}-1}$

c) $\frac{\sqrt{3}-1}{2\sqrt{2}}$

d) $\frac{\sqrt{3}-1}{\sqrt{3}+1}$

✓ ANSWER: (C)

$\cos 75° = \cos(45° + 30°)$

$= \cos 45° \cdot \cos 30° - \sin 45° \cdot \sin 30°$

$= \left(\frac{1}{\sqrt{2}}\right)\left(\frac{\sqrt{3}}{2}\right) - \left(\frac{1}{\sqrt{2}}\right)\left(\frac{1}{2}\right)$

$= \frac{\sqrt{3}-1}{2\sqrt{2}}$

Q.12

TRIGONOMETRY

If a ladder is inclined to the wall with base angle 30°, then the angle in radians is

a) $\frac{\pi}{3}$

b) $\frac{\pi}{4}$

c) $\frac{\pi}{2}$

d) $\frac{\pi}{6}$

✓ ANSWER: (D)

Angle in radian = $\left(\frac{\pi}{180°}\right) \times 30° = \frac{\pi}{6}$ radian

Q.13

TRIGONOMETRY

If $\sin A = \frac{3}{5}$ and $\cos B = \frac{12}{13}$, then $\cos(A - B)$ is

a) $-\frac{33}{65}$

b) $\frac{63}{65}$

c) $-\frac{63}{65}$

d) $\frac{33}{65}$

✓ ANSWER: (B)

$\sin A = \frac{3}{5} \rightarrow \cos A = \frac{4}{5}$ (using 3-4-5 triplet)

$\cos B = \frac{12}{13} \rightarrow \sin B = \frac{5}{13}$ (using 5-12-13 triplet)

$\cos(A-B) = \cos A \cdot \cos B + \sin A \cdot \sin B$

$= \left(\frac{4}{5}\right)\left(\frac{12}{13}\right) + \left(\frac{3}{5}\right)\left(\frac{5}{13}\right)$

$= \frac{48}{65} + \frac{15}{65} = \frac{63}{65}$

Q.14

DIFFERENTIAL CALCULUS

If $y = 2 \sin x - 4 \cot x + e^{-x} + 7$, then dy/dx is

a) $-2 \cos x - 4 \operatorname{cosec}^2 x - e^{-x} + 7x$

b) $2 \cos x + 4 \operatorname{cosec}^2 x - e^{-x}$

c) $2 \cos x + 4 \operatorname{cosec}^2 x + e^{x} + 7$

d) $2 \sin x - 4 \operatorname{cosec}^2 x + e^{x}$

✓ ANSWER: (B)

$dy/dx = 2 \cos x - 4(-\operatorname{cosec}^2 x) - e^{-x}$

$= 2 \cos x + 4 \operatorname{cosec}^2 x - e^{-x}$

Q.15

DIFFERENTIAL CALCULUS

If $y = (x + \frac{1}{x})^2$, then dy/dx is

a) $x^2 + \frac{1}{x^2} + 2$

b) $2x + \frac{1}{x^2}$

c) $2x - \frac{2}{x^3}$

d) $2x + \frac{1}{x}$

✓ ANSWER: (C)

$y = (x + \frac{1}{x})^2 = x^2 + \frac{1}{x^2} + 2$

$dy/dx = 2x - \frac{2}{x^3}$

Q.16

DIFFERENTIAL CALCULUS

If $y = e^{3x} + e^{7x}$, then d^2y/dx^2 at $x = 0$ is

a) 10

b) 58

c) 1

d) 2

✓ ANSWER: (B)

$dy/dx = 3e^{3x} + 7e^{7x}$

$d^2y/dx^2 = 9e^{3x} + 49e^{7x}$

At $x = 0$: $9(1) + 49(1) = 58$

Q.17

DIFFERENTIAL CALCULUS

If y = log(sec x + tan x), then dy/dx is

a) sec x + tan x

b) $\dfrac{1}{\sec x + \tan x}$

c) sec x

d) $\sec^2 x$

✓ ANSWER: (C)

$dy/dx = [\sec x \tan x + \sec^2 x] / (\sec x + \tan x)$

$= \sec x(\tan x + \sec x) / (\sec x + \tan x)$

$= \sec x$

Q.18

INTEGRAL CALCULUS

$\int(3x^2 + 2x + 2)^{3/2}(3x + 1)\, dx$ is

a) $\dfrac{3x + 5}{5} + C$

b) $\dfrac{3x - 5}{5} + C$

c) $\left(\dfrac{3x^2 - 2x - 2}{5}\right)^{5/2} + C$

d) $\dfrac{(3x^2 + 2x + 2)^{5/2}}{5} + C$

✓ ANSWER: (D)

Let $f(x) = 3x^2 + 2x + 2$, $f'(x) = 6x + 2 = 2(3x+1)$

$\int f(x)^{3/2}(3x+1)dx = \dfrac{1}{2}\int f(x)^{3/2} \cdot f'(x)dx$

$= \dfrac{1}{2} \cdot \dfrac{f(x)^{5/2}}{5/2} + C$

$= \dfrac{(3x^2+2x+2)^{5/2}}{5} + C$

Q.19

INTEGRAL CALCULUS

The integral of $x^3 + \frac{1}{x} + e^x$ is

a) $\frac{x^4}{4} + \log x - e^x + C$

b) $\frac{x^4}{4} + \log x + e^x + C$

c) $3x^2 - \frac{1}{x^2} + e^x + C$

d) $3x^2 - \frac{1}{x^2} - e^x + C$

✓ ANSWER: (B)

$$\int(x^3 + \frac{1}{x} + e^x)dx = \frac{x^4}{4} + \log x + e^x + C$$

Q.20

INTEGRAL CALCULUS

$\int x \sin x \, dx =$

a) $x \cos x - \sin x + C$

b) $-x \cos x + \sin x + C$

c) $x \cos x + \sin x + C$

d) $-x \cos x - \sin x + C$

✓ ANSWER: (B)

Using integration by parts: $\int u \, dv = uv - \int v \, du$

$u = x$, $dv = \sin x \, dx$

$= x(-\cos x) - \int(-\cos x)dx$

$= -x \cos x + \sin x + C$

02

CHAPTER TWO

Project Management Skills

20 Marks

Questions 21 – 40

Q.21

PM BASICS

Which of the following are not obstacles in project management?

a) Project complexities
b) Project risks
c) Changes in technology
d) Early identification of problems

✓ ANSWER: (D)

Early identification of problems is a benefit of project management, not an obstacle.

Q.22

CONSULTANTS

__________ provides guidance as well as direction to the project and he/she is not a part of the project team.

a) CEO
b) Project Manager
c) Team Leader
d) Project Consultant

✓ ANSWER: (D)

Project Consultant guides the project but is not a part of the project team.

Q.23

PROJECT TYPES

Repair of dam in case of damage due to natural calamities is an example of

a) Normal project
b) Crash project
c) Risky project
d) Disaster project

✓ ANSWER: (D)

A project performed due to the effect of natural calamities is known as a **disaster project**.

Q.24

OBJECTIVES

__________ refers to a detailed description of the expected outcome of a project.

a) Project scope

b) Project operation

c) Project objective

d) Project process

✓ ANSWER: (C)

Project objective illustrates the tasks you execute to achieve the result.

Q.25

WBS

The process of dividing the project into number of tasks in order to carry out effective project administration is:

a) PPM

b) WBS

c) PEP

d) Project diary

✓ ANSWER: (B)

Work Breakdown Structure (WBS) is the process of dividing project work into tasks.

Q.26

ADMINISTRATION

The project __________ relieves a project team from most regular work such as planning, tracking and reporting responsibility.

a) Manager

b) CEO

c) Administration

d) Leader

✓ ANSWER: (C)

Project **administration** relieves a project team from regular work.

Q.27

DIRECTION

The use of authority to channelize the activities of the project on desired lines is referred to as

a) Project direction

b) Project coordination

c) Project communication

d) Project execution

✓ ANSWER: (A)

Project direction channelizes the activities of the project on desired lines.

Q.28

WBS

__________ illustrates total project work into divisions of major work packages.

a) PERT
b) CPM
c) WBS
d) PEP

✓ ANSWER: (C)

WBS divides the work into major packages.

Q.29

OVERRUNS

__________ is a condition wherein a construction or manufacturing project does not complete well within its prescribed time schedule.

a) Time analysis
b) Time estimate
c) Time overrun
d) Cost overrun

✓ ANSWER: (C)

Time overrun is when a project cannot be completed within its prescribed time schedule.

Q.30

RISK

A project __________ creates damages or adverse effects to either project activities or project deliverables.

a) Failure
b) Undertaken
c) Risk
d) Analysis

✓ ANSWER: (C)

Damages or uncertainties affecting a project is known as **risk**.

Q.31

RISK ANALYSIS

Identify the correct risk analysis which has typically one variable changed at a time

a) Sensitivity Analysis
b) Scenario Analysis
c) Best and Worst Case Analysis
d) Simulation Analysis

✓ ANSWER: (A)

Sensitivity Analysis is a type of risk analysis done by changing one variable at a time.

Q.32

RISK

Risk usually __________ as the project progresses.

a) Increases

b) Reduces

c) Remains same

d) Becomes negligible

✓ ANSWER: (B)

As a project progresses towards its goal, risk usually **reduces**.

Q.33

SMART

In 'SMART', the letter R stands for __________.

a) Realistic

b) Reliable

c) Reasonable

d) Relation

✓ ANSWER: (A)

SMART: S-Specific, M-Measurable, A-Achievable, R- **Realistic** , T-Timely

Q.34

EVALUATION

__________ is a step-by-step process of collecting, recording and organising information about project results.

a) Project planning

b) Project evaluation

c) Project scheduling

d) Project monitoring

✓ ANSWER: (B)

Project evaluation is a process where information is collected, recorded, and organised to check project results.

Q.35

POLICIES

Which one of the following is not a policy of a project?

a) Extent of work given to outside contractors

b) Number of contractors to be employed

c) Monitoring the progress of project

d) Terms of contract

✓ ANSWER: (C)

Monitoring the progress is a phase of project life cycle, not a policy.

Q.36

SCHEDULING

In scheduling, 'EST' stands for:

a) Easiest Start Time

b) Earliest Start Time

c) Earliest Schedule Time

d) Earliest Schedule Technique

✓ ANSWER: (B)

EST stands for **Earliest Start Time**.

Q.37

AUDITOR

___________ is an expert in measuring, confirming, investigating and reporting status of a project.

a) Project Guide

b) Project Reviewer

c) Project Auditor

d) Project Advisor

✓ ANSWER: (C)

Project Auditor is the one who reports the status of a project.

Q.38

TIME ESTIMATES

___________ is the time estimate if there are no hurdles in an activity.

a) Pessimistic time

b) Optimistic time

c) Most likely time

d) Expected estimated time

✓ ANSWER: (B)

Optimistic time is where project time is scheduled based on quickest completion without any risk.

Q.39

LIFE CYCLE

___________ is the last phase of the project life cycle which reports overall achievements of a project in terms of defined performance measures.

a) Project closure

b) Project initiation

c) Project execution

d) Project planning

✓ ANSWER: (A)

Project closure is the final phase of the project life cycle.

Q.40

NETWORK ANALYSIS

The objective of network analysis in a project is to

a) Minimise total project duration

b) Minimise total project cost

c) Minimise production delays

d) Minimise manpower requirements

✓ ANSWER: (A)

Minimise total project duration is the objective of network analysis.

03

CHAPTER THREE

IT Skills

20 Marks

Questions 41 – 60

Q.41

PSEUDOCODE

What is the output of the following?

```
a = 10
if (a > 1) then
  print "TWENTY"
else
  print "TEN"
```

a) 10

b) TEN

c) TWENTY

d) No Output

✓ ANSWER: (C)

Since a = 10, and 10 > 1 is true, the output is **TWENTY**.

Q.42

FLOWCHARTS

The symbol used to connect two parts of a big flowchart is

a) Oval

b) Rectangle

c) Diamond

d) Circle

✓ ANSWER: (D)

In flowcharts, **circles** are used as connector symbols to show the continuation of flow across multiple pages.

Q.43

FLOWCHARTS

Which symbol is used to represent start or stop in a flowchart?

a) Rectangle

b) Diamond

c) Oval

d) Circle

✓ ANSWER: (C)

The **oval** symbol represents the start and end points of a process.

Q.44

CSS

What does CSS stand for?

a) Colourful Style Sheet

b) Cascading Style Sheet

c) Creative Style Sheet

d) Computer Style Sheet

ANSWER: (B)

CSS stands for **Cascading Style Sheet**.

Q.45

WEB

Which of the following is not a web browser?

a) Google Chrome

b) Netscape Navigator

c) Safari

d) Apache

ANSWER: (D)

Apache is a web server software, not a web browser.

Q.46

CSS

What is the correct syntax to change the background colour with CSS?

a) background-color = "purple";

b) background-color : purple;

c) background-color = orange

d) change-background-color : purple

ANSWER: (B)

In CSS, properties are separated from values by a colon. `background-color: purple;`

Q.47 HTML

Which HTML tag gives a scrolling effect of a given text?

a) <div>

b) <scroll>

c) <marquee>

d) <effect>

✓ ANSWER: (C)

The `<marquee>` tag is used to create scrolling text in HTML.

Q.48 ERP

ERP package will handle __________ business functionality/functionalities.

a) One

b) Two

c) Three

d) Multiple

✓ ANSWER: (D)

ERP integrates all functional units — it handles **multiple** business functionalities.

Q.49 ORGANIZATION

Which one is a type of organizational structure?

a) Responsibility structure

b) Behavioural structure

c) Functional structure

d) Relational structure

✓ ANSWER: (C)

Functional structure is a type of organisational structure based on functional units.

Q.50 USE CASE

The __________ describes how an actor uses a system to accomplish a particular goal for describing the activities.

a) Process case

b) Work case

c) Use case

d) Sub process

✓ ANSWER: (C)

Use case describes how an employee uses a system.

Q.51

ERP

Which of the following is not a characteristic of First-generation ERP system?

a) They were based on mainframe computer

b) They focused on accounting and finance

c) They used centralized database

d) They supported real-time data processing

✓ ANSWER: (D)

First-generation ERP systems were standalone applications for specific functions. **Real-time data processing** was not a characteristic.

Q.52

CLOUD

Which cloud service model enables users to develop, run, manage application without worrying about underlying infrastructure?

a) IaaS

b) PaaS

c) SaaS

d) NaaS

✓ ANSWER: (B)

PaaS (Platform as a Service) provides a platform for developing and managing applications without infrastructure concerns.

Q.53

GOOGLE APPS

Which of the following is not a Google application?

a) Google Docs

b) Google Slides

c) Google Sheets

d) Google Word

✓ ANSWER: (D)

Google Word does not exist. Google's word processing application is called Google Docs.

Q.54 IOT

IoT is a __________.

a) Network of sensors

b) Network of objects in a ring structure

c) Network of virtual objects

d) Network of physical objects embedded with sensors

✓ ANSWER: (D)

IoT refers to the network of physical objects embedded with sensors, software, and other technologies for connecting and exchanging data.

Q.55 GOOGLE APPS

__________ is a file storage and synchronization service developed by Google which provides complete office tools with cloud storage.

a) Google Colab

b) Google Drive

c) Google Meet

d) AWS Cloud

✓ ANSWER: (B)

Google Drive is a file storage and synchronization service with cloud storage and office tools integration.

Q.56 CYBER SECURITY

Which of the following is not a security issue in a personal computer?

a) Ransomware Attack

b) Data Breach

c) Code Injection

d) IoT

✓ ANSWER: (D)

IoT is a technology concept, not a security issue in personal computers.

Q.57 FIREWALL

Which of the following is not a firewall?

a) Packet filtering

b) Stateful inspection

c) Panda

d) Circuit Level Gateway

✓ ANSWER: (C)

Panda is a cybersecurity brand (antivirus software), not a type of firewall.

Q.58 NETWORKING

"S" in HTTPS stands for __________.

a) Safe

b) Secure

c) System

d) Server

✓ ANSWER: (B)

HTTPS = HyperText Transfer Protocol **Secure**.

Q.59 HASHING

The process used to get an output 43205 from a given input "DCTE" is

a) Hashing

b) Compression

c) Decompression

d) Encryption

✓ ANSWER: (A)

Hashing converts an input of any size into a fixed-length alphanumeric string.

Q.60

PROGRAMMING

a = 10, b = 20, a = a + b, b = a − b, a = a − b. The final values of a and b are:

a) a=30, b=10

b) a=30, b=20

c) a=20, b=10

d) a=20, b=30

✓ ANSWER: (C)

```
a=10, b=20
a = a+b = 30
b = a−b = 30−20 = 10
a = a−b = 30−10 = 20
∴ a=20, b=10 (values are swapped)
```

04

CHAPTER FOUR

Statistics & Analytics

20 Marks

Questions 61 – 80

Q.61

QUESTIONNAIRES

Which one of the following is not essential for good questionnaires?

a) Logically arranged

b) Avoiding personal questions

c) Limiting number of questions

d) Unlimited number of questions

✓ **ANSWER: (D)**

Unlimited number of questions is not essential for a good questionnaire.

Q.62

FREQUENCY

Both categorical and numerical data variables can be used for

a) Continuous variable

b) Class intervals

c) Distribution of frequency

d) Frequency distribution table

✓ **ANSWER: (D)**

Frequency distribution table can be used for both categorical and numerical data.

Q.63

EXCEL

Match the following:

Column A	Column B
a) Mean	i) $Q_3 - Q_1$
b) Absolute value	ii) =AVERAGE(array)
c) IQR	iii) =ABS(cell)

a) a-i, b-ii, c-iii

b) a-ii, b-iii, c-i

c) a-ii, b-i, c-iii

d) a-i, b-iii, c-ii

✓ **ANSWER: (B)**

Mean → =AVERAGE(array), Absolute value → =ABS(cell), IQR → $Q_3 - Q_1$

Q.64

CHARTS

___________ type of chart is used to show the relationship between numerical and categorical variable

a) Plot

b) Pie

c) Bar

d) Line

✓ ANSWER: (C)

Bar chart shows the relationship between numerical and categorical variables.

Q.65

PYTHON

Who developed Python language?

a) Zim Den

b) Guido van Rossum

c) Niene Stom

d) Wick van Rossum

✓ ANSWER: (B)

Python was created by **Guido van Rossum**.

Q.66

DATA CLEANING

Which one of the following is not the benefit of data cleaning?

a) Removal of errors when multiple sources of data are at play

b) Boosts results and revenue

c) Helps with faster decision-making and more efficient business practices

d) Consumes more time

✓ ANSWER: (D)

Consuming more time is not a benefit of data cleaning.

Q.67

PYTHON

Is Python code compiled or interpreted?

a) Python code is both compiled and interpreted

b) Python code is neither compiled nor interpreted

c) Python code is only compiled

d) Python code is only interpreted

✓ ANSWER: (A)

Python code is first compiled into bytecode, then interpreted by the Python Virtual Machine (PVM).

Q.68

INTERVIEWS

Which one of the following is not a type of interview?

a) Unstructured

b) Semi-structured

c) Audio recorder

d) Structured

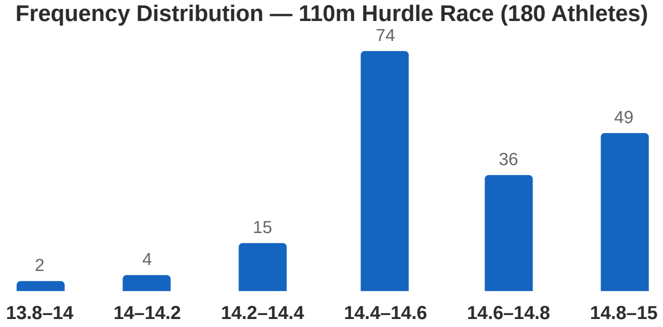
✓ ANSWER: (C)

Audio recorder is not a type of interview.

Q.69

DATA INTERPRETATION

The time taken by 180 athletes to run a 110m hurdle race is tabulated. The number of athletes who completed the race in less than 14.6 seconds is:



a) 19
b) 95
c) 131
d) 180

✓ ANSWER: (B)

```
Athletes who completed in < 14.6s:
= 2 + 4 + 15 + 74 = 95
```

Q.70

RANGE

The higher value of the range represents:

a) greater spread of the Data
b) lesser spread of the Data
c) equal to range of the Data
d) average of the Data

✓ ANSWER: (A)

A higher range value indicates **greater spread** of data.

Q.71

CENTRAL TENDENCY

The middle number in a given data set is called:

a) Mean
b) Mode
c) Frequency
d) Median

✓ ANSWER: (D)

Median is the middle-most value in arranged data.

Q.72

DATA TYPES

__________ is the example of qualitative data.

a) Temperature

b) Ability of students

c) Prices

d) Area

✓ ANSWER: (B)

Ability of students is not numerically measurable — it is qualitative data.

Q.73

DATA COLLECTION

Direct personal observation is a method of collection of:

a) Secondary Data

b) Primary Data

c) Mixed Data

d) Unpublished Data

✓ ANSWER: (B)

Direct personal observation is a method of collecting primary data.

Q.74

EXCEL

The in-built function to find the mean of a given data in Excel is:

a) =AVERAGE()

b) =MEAN()

c) =MEDIAN()

d) =MODE()

✓ ANSWER: (A)

Excel formula for mean is `=AVERAGE()`

Q.75

FREQUENCY

The formula for Relative Frequency is:

a) Frequency × No. of observations

b) $\frac{\text{No. of observations}}{\text{Frequency}}$

c) $\frac{\text{Frequency}}{\text{No. of observations}}$

d) No. of observations + Frequency

✓ ANSWER: (C)

Relative frequency = $\frac{f}{n}$

Q.76 PYTHON

In Python, "str" is a:

a) Numeric type
b) Binary type
c) Text type
d) Sequence type

✓ ANSWER: (C)

str is the Text type in Python.

Q.77 PYTHON

The results of Python program get displayed in

a) IDLE 3.1.9 Window
b) IELD shell 3.9.1 Window
c) IDLE shell 3.9.1 Window
d) ILDE shell 3.1.9 Window

✓ ANSWER: (C)

IDLE (Integrated Development and Learning Environment) shell 3.9.1 Window.

Q.78 DATA CLEANING

The process of Data cleaning is

a) Removing virus
b) Deduplication
c) Survey
d) Google form

✓ ANSWER: (B)

Data cleaning involves removing duplicate data, called deduplication.

Q.79 DATA TYPES

Data collected for the first time is called:

a) Published Data
b) Mixed Data
c) Primary Data
d) Secondary Data

✓ ANSWER: (C)

Data collected for the first time directly from the field by an investigator is called primary data.

Q.80

EXCEL

The Excel formula for standard deviation of a sample is:

a) STDEV.P

b) SDTEV.S

c) STDEV.S

d) SDTEV.P

✓ ANSWER: (C)

Excel formula for sample standard deviation is `=STDEV.S`

05

CHAPTER FIVE

Fundamentals of Electrical & Electronics Engineering

20 Marks

Questions 81 – 100

Q.81

SAFETY

CPR stands for

- **a)** Cardio Permanent Respiration
- **b)** Cardio Pulmonary Respiration
- **c)** Cardio Pulmonary Resuscitation
- **d)** Cardio Pulmonary Retrieval

✓ ANSWER: (C)

CPR stands for **Cardio Pulmonary Resuscitation**.

Q.82

CIRCUITS

Two 5Ω resistors are connected in parallel and this combination is connected in series with a 2.5Ω resistor. The effective resistance is:

- **a)** 0.5 Ω
- **b)** 5 Ω
- **c)** 2.5 Ω
- **d)** 7.5 Ω

✓ ANSWER: (B)

$$Rp = \frac{5\times5}{5+5} = \frac{25}{10} = 2.5\ \Omega$$

$$\text{Total} = Rp + 2.5 = 2.5 + 2.5 = 5\ \Omega$$

Q.83

OHM'S LAW

The relationship between Voltage (V), Current (I) and Resistance (R) is:

- **a)** I = VR
- **b)** R = VI
- **c)** V = IR
- **d)** V = I/R

✓ ANSWER: (C)

Ohm's Law: `V = IR`

Q.84

PROTECTION

ELCB stands for:

a) Earth Line Circuit Breaker

b) Electric Load Circuit Breaker

c) Earth Leakage Circuit Breaker

d) Electric Line Circuit Breaker

✓ ANSWER: (C)

ELCB stands for **Earth Leakage Circuit Breaker**.

Q.85

FUSE

Good fuse wire must have:

a) Low melting point

b) High melting point

c) High resistivity

d) High melting point and high resistivity

✓ ANSWER: (A)

A good fuse wire must have **low melting point** to break the circuit quickly on overload.

Q.86

MOTORS

A 10 HP submersible pump set is used in a borewell for irrigation purpose. It normally uses ___________ type of motor.

a) Single-phase induction motor

b) Three-phase induction motor

c) DC motor

d) FHP motor

✓ ANSWER: (B)

A **three-phase induction motor** is used in borewells for irrigation with 10 HP pump sets.

Q.87

BATTERIES

SMF in batteries stands for:

a) Sealed Moulded - Free

b) Semiconductor Maintenance - Free

c) Sealed Maintenance - Free

d) Solar Maintenance - Free

✓ ANSWER: (C)

SMF stands for **Sealed Maintenance Free** batteries.

Q.88

SENSORS

The typical use of LVDT is to sense

a) Light intensity

b) Humidity

c) Pressure

d) Temperature

✓ ANSWER: (C)

LVDT sensor is used to sense displacement, force, and **pressure**.

Q.89

ELECTRONICS

A p-n junction diode is a

a) Bidirectional switch

b) Unidirectional switch

c) Bidirectional Controlled Switch

d) Unidirectional Controlled Switch

✓ ANSWER: (B)

A P-N junction diode is a **unidirectional switch** — it allows current flow only from anode to cathode.

Q.90

RECTIFIERS

The purpose of C Filter in a power supply is to

a) Degrade regulation

b) Reduce ripple

c) Reduce gain

d) Reduce the chance of electric shock

✓ ANSWER: (B)

The C filter is used to decrease unwanted **ripple voltage**. It is connected in parallel with the load.

Q.91

BATTERIES

Lead acid cell is an example of

a) Secondary cell

b) Solar cell

c) Primary cell

d) Fuel cell

✓ ANSWER: (A)

A **secondary cell** is designed to be recharged and reused. Examples: Lead-Acid, Nickel Cadmium, Lithium ion.

Q.92

AC THEORY

Time taken by the alternating quantity to complete one cycle is called

a) Frequency

b) Time period

c) Amplitude

d) Wavelength

✓ ANSWER: (B)

The time taken to complete one cycle is called **time period**.

Q.93

AUTOMATION

What does "PLC" stand for in the context of Industrial Automation?

a) Programmable Logic Circuit

b) Process Logic Controller

c) Programmable Logic Controller

d) Processor Logic Controller

✓ ANSWER: (C)

PLC stands for **Programmable Logic Controller** used for industrial automation.

Q.94

AC THEORY

Maximum value attained by the alternating quantity either in positive half-cycle or negative half-cycle is called

a) Frequency

b) Altitude

c) Amplitude

d) Wavelength

✓ ANSWER: (C)

Amplitude is the maximum value attained by an alternating quantity in either half cycle.

Q.95

CIRCUITS

Which of the following statement is correct?

a) In series circuit the current through each resistor is different

b) In parallel circuit the voltage across each resistor is same

c) In parallel circuit the equivalent resistance is sum of individual resistances

d) In series circuit the reciprocal of equivalent resistance is sum of reciprocal of individual resistances

✓ ANSWER: (B)

In a parallel circuit, the **voltage remains constant** across each resistor.

Q.96

BATTERIES

The cell which cannot be recharged and reused is

a) Secondary cell

b) Battery

c) Primary cell

d) Simple cell

✓ ANSWER: (C)

A **primary cell** is designed to be used once and discarded; it cannot be recharged.

Q.97

BATTERIES

Specific gravity of a fully charged lead acid battery will be in the range of

a) 1000 to 1210

b) 1100 to 1310

c) 1150 to 1310

d) 1230 to 1300

✓ ANSWER: (D)

Specific gravity of a fully charged lead acid battery is in the range of **1230 to 1300**.

Q.98

SENSORS

The best sensor for sensing vibration is

a) Thermistor

b) Thermocouple

c) Piezoelectric crystal

d) Hygrometer

✓ ANSWER: (C)

The **piezoelectric crystal** sensor is used for sensing vibration.

Q.99

BATTERIES

During charging of batteries

a) Chemical → Mechanical energy

b) Mechanical → Chemical energy

c) Electrical → Chemical energy

d) Chemical → Electrical energy

✓ ANSWER: (C)

During charging, **electrical energy is converted to chemical energy**.

Q.100

ENERGY

When an electric stove is connected across 230 volts, 50 Hz AC supply for one hour and takes a current of 10 amperes, the energy consumed is:

a) 230 units

b) 2.3 units

c) 23 units

d) 0.23 units

✓ ANSWER: (B)

$$\text{Energy} = V \times I \times t = 230 \times 10 \times 1 = 2300 \text{ Wh}$$

$$= \frac{2300}{1000} = 2.3 \text{ units (kWh)}$$

2025

PART THREE

DCET 2025

100 Questions • 5 Subjects • 100 Marks

Previous Year Question Paper & Complete Solutions

01

CHAPTER ONE

Engineering Mathematics

20 Marks

Questions 1 – 20

Q.01 MATRICES

Choose the correct answer from the following:

a) Every identity matrix is a scalar matrix

b) Every scalar matrix is an identity matrix

c) Every diagonal matrix is an identity matrix

d) A square matrix whose each element is one is an identity matrix

✓ ANSWER: (A)

An identity matrix has 1's on the diagonal and 0's elsewhere. A scalar matrix has equal diagonal elements and 0's elsewhere. So every identity matrix is a scalar matrix (with scalar = 1).

Q.02 DETERMINANTS

The value of $\begin{vmatrix} 3 & 1 & -9 \\ -1 & 2 & 3 \\ 2 & 1 & -6 \end{vmatrix}$ is

a) 45

b) 90

c) 0

d) −45

✓ ANSWER: (C)

```
= 3(−12−3) −1(6−6) +(−9)(−1−4)
= 3(−15) −1(0) −9(−5)
= −45 − 0 + 45 = 0
```

Q.03

EIGENVALUES

The characteristic equation of A = $\begin{bmatrix} 1 & -2 \\ 3 & 4 \end{bmatrix}$ is

a) $\lambda^2 - 3\lambda - 2 = 0$

b) $\lambda^2 - 5\lambda + 10 = 0$

c) $\lambda^2 - 5\lambda - 2 = 0$

d) $\lambda^2 + 3\lambda + 10 = 0$

✓ ANSWER: (B)

$|A - \lambda I| = 0$

$|(1-\lambda)(4-\lambda) - (-2)(3)| = 0$

$4 - \lambda - 4\lambda + \lambda^2 + 6 = 0$

$\lambda^2 - 5\lambda + 10 = 0$

Q.04

MATRICES

If |A|I = $\begin{bmatrix} -3 & 0 \\ 0 & -3 \end{bmatrix}$, then A·adj(A) is equal to

a) $\begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix}$

b) $\begin{bmatrix} 9 & 0 \\ 0 & 9 \end{bmatrix}$

c) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

d) $\begin{bmatrix} -3 & 0 \\ 0 & -3 \end{bmatrix}$

✓ ANSWER: (D)

For any square matrix, $\boxed{A \cdot adj(A) = |A| \cdot I}$. Given |A|I = $\begin{bmatrix} -3 & 0 \\ 0 & -3 \end{bmatrix}$, so A·adj(A) = $\begin{bmatrix} -3 & 0 \\ 0 & -3 \end{bmatrix}$.

Q.05

STRAIGHT LINES

The slope of the straight line joining two points (3, −4) and (−1, 5) is

a) $\frac{4}{9}$

b) $-\frac{4}{9}$

c) $-\frac{1}{4}$

d) $-\frac{9}{4}$

✓ ANSWER: (D)

$$m = \frac{y_2-y_1}{x_2-x_1} = \frac{5-(-4)}{-1-3} = \frac{9}{-4} = -\frac{9}{4}$$

Q.06

STRAIGHT LINES

The angle of inclination of the straight line is π/4 and y-intercept is 3, then equation of the straight line is

a) $y = x + 3$

b) $y = x - 3$

c) $y = -x - 3$

d) $y = -x + 3$

✓ ANSWER: (A)

$$\theta = \frac{\pi}{4} \rightarrow m = \tan\left(\frac{\pi}{4}\right) = 1,\ c = 3$$

$$y = mx + c = 1\cdot x + 3 = x + 3$$

Q.07

STRAIGHT LINES

The equation of straight line passing through (−5, 4) and parallel to X-axis is

a) $x - y + 9 = 0$

b) $x + 5 = 0$

c) $x + y + 1 = 0$

d) $y - 4 = 0$

✓ ANSWER: (D)

A line parallel to X-axis has slope m = 0. Passing through (−5, 4): $\boxed{y - 4 = 0}$

Q.08

STRAIGHT LINES

What is the equation of the straight line passing through the points (1, 5) and (3, 2)?

a) $7x - 4y + 13 = 0$

b) $3x + 2y - 17 = 0$

c) $3x + 2y - 13 = 0$

d) $3x + 2y + 13 = 0$

✓ ANSWER: (C)

```
m = (2−5)/(3−1) = −3/2
y − 5 = (−3/2)(x − 1)
2y − 10 = −3x + 3 → 3x + 2y − 13 = 0
```

Q.09

TRIGONOMETRY

The ramp in front of a gate is elevated by an angle of $\frac{\pi}{3}$ radians, then the angle in degrees is

a) 15°

b) 75°

c) 60°

d) 30°

✓ ANSWER: (C)

$$\text{Degrees} = \text{Radians} \times \frac{180}{\pi} = \left(\frac{\pi}{3}\right) \times \left(\frac{180}{\pi}\right) = 60^\circ$$

Q.10

TRIGONOMETRY

If $\sin(90 - \theta) + \cos\theta = \sqrt{2}\cos(90 - \theta)$, then the value of $\operatorname{cosec}\theta$ is

a) $\frac{\sqrt{3}}{2}$

b) $\sqrt{\frac{3}{2}}$

c) $\sqrt{\frac{2}{3}}$

d) $\sqrt{3}$

✓ **ANSWER: (B)**

$\cos\theta + \cos\theta = \sqrt{2}\sin\theta$

$2\cos\theta = \sqrt{2}\sin\theta \rightarrow \tan\theta = \frac{2}{\sqrt{2}} = \sqrt{2}$

Opp = $\sqrt{2}$, Adj = 1, Hyp = $\sqrt{3}$

$\operatorname{cosec}\theta = \frac{Hyp}{Opp} = \frac{\sqrt{3}}{\sqrt{2}} = \sqrt{\frac{3}{2}}$

Q.11

TRIGONOMETRY

If A, B, C are the angles of a triangle, then the incorrect relation is

a) $\sin\left(\frac{A+B}{2}\right) = \cos\left(\frac{C}{2}\right)$

b) $\cos\left(\frac{A+B}{2}\right) = \sin\left(\frac{C}{2}\right)$

c) $\tan\left(\frac{A+B}{2}\right) = \sin\left(\frac{C}{2}\right)$

d) $\cot\left(\frac{A+B}{2}\right) = \tan\left(\frac{C}{2}\right)$

✓ **ANSWER: (C)**

Since A+B+C = 180°, $\frac{A+B}{2} = 90° - \frac{C}{2}$. So $\tan\left(\frac{A+B}{2}\right)$ should equal $\cot\left(\frac{C}{2}\right)$, not $\sin\left(\frac{C}{2}\right)$. Option (c) is incorrect.

Q.12

TRIGONOMETRY

The value of sin 40° + sin 20° is

a) $\sqrt{3}\cos 10°$

b) $\sqrt{3}\cos 20°$

c) $\cos 10°$

d) $\sqrt{2}\cos 10°$

✓ ANSWER: (C)

$$\sin C + \sin D = 2\sin\left(\frac{C+D}{2}\right)\cos\left(\frac{C-D}{2}\right)$$

$$= 2\sin(30°)\cos(10°)$$

$$= 2 \times \left(\frac{1}{2}\right) \times \cos 10° = \cos 10°$$

Q.13

DIFFERENTIAL CALCULUS

The derivative of $y = \frac{1+x}{1-x}$ is

a) $\frac{2}{(1-x)^2}$

b) $\frac{-2}{(1-x)^2}$

c) $\frac{2}{1-x^2}$

d) $\frac{-2x}{(1-x)^2}$

✓ ANSWER: (A)

$$dy/dx = \frac{(1-x)(1) - (1+x)(-1)}{(1-x)^2}$$

$$= \frac{1-x+1+x}{(1-x)^2} = \frac{2}{(1-x)^2}$$

Q.14

DIFFERENTIAL CALCULUS

If $y = \log(\sqrt{\tan x})$, then the value of dy/dx at $x = \frac{\pi}{4}$ is

a) ∞

b) 1

c) 0

d) $\frac{1}{2}$

ANSWER: (B)

$$\frac{dy}{dx} = \left(\frac{1}{\sqrt{\tan x}}\right) \times \left(\frac{1}{2\sqrt{\tan x}}\right) \times \sec^2 x$$

$$= \frac{\sec^2 x}{2\tan x}$$

$$\text{At } x=\frac{\pi}{4}: \quad \frac{(\sqrt{2})^2}{2 \cdot 1} = \frac{2}{2} = 1$$

Q.15

DIFFERENTIAL CALCULUS

If $y = (2 + 3\log x)^{10}$, then dy/dx at x = 1 is

a) $15(2^9)$

b) $(15 \times 2)^{10}$

c) $15(2^{10})$

d) $10(5^9)$

ANSWER: (C)

$$\frac{dy}{dx} = 10(2+3\log x)^9 \times \left(\frac{3}{x}\right)$$

$$\text{At } x=1\ (\log 1=0):\ = 10(2)^9 \times 3 = 30(2^9)$$

$$= 15 \times 2 \times 2^9 = 15(2^{10})$$

Q.16

IMPLICIT DIFFERENTIATION

If $y = \sqrt{(\sin x + y)}$, then dy/dx =

a) $\dfrac{\cos x}{2y - 1}$

b) $\dfrac{\cos x}{1 - 2y}$

c) $\dfrac{\sin x}{2y - 1}$

d) $\dfrac{\sin x}{1 - 2y}$

ANSWER: (A)

$y^2 = \sin x + y$ (squaring both sides)

$2y(dy/dx) = \cos x + dy/dx$

$dy/dx(2y - 1) = \cos x$

$dy/dx = \dfrac{\cos x}{2y - 1}$

Q.17

INTEGRAL CALCULUS

$\int \cot^2 x \, dx$ =

a) $-\cot x - x + c$

b) $\cot x - x + c$

c) $\tan x - x + c$

d) $\tan^2 x + c$

ANSWER: (A)

$\cot^2 x = \mathrm{cosec}^2 x - 1$

$\int(\mathrm{cosec}^2 x - 1)dx = -\cot x - x + c$

Q.18

DEFINITE INTEGRAL

The value of $\int_0^{\pi/4} \frac{\sec^2 x}{1+\tan x}\, dx$ is

a) log2 + 1

b) log1

c) log 2

d) log2 − 1

✓ ANSWER: (C)

Let $f(x) = 1+\tan x$, $f'(x) = \sec^2 x$

$\int \frac{f'(x)}{f(x)}\, dx = \log|f(x)|$

$= [\log(1+\tan x)]_0^{\frac{\pi}{4}}$

$= \log(1+1) - \log(1+0) = \log 2 - 0 = \log 2$

Q.19

INTEGRAL CALCULUS

$\int\sqrt{(1 + \cos 2x)}\, dx$ is

a) 2cosx + c

b) 2sinx + c

c) $\sqrt{2}$ cosx + c

d) $\sqrt{2}$ sinx + c

✓ ANSWER: (D)

$1 + \cos 2x = 2\cos^2 x$

$\sqrt{(2\cos^2 x)} = \sqrt{2} \cos x$

$\int\sqrt{2} \cos x\, dx = \sqrt{2} \sin x + c$

Q.20

The value of $\int_0^{\pi/4} \sin x \cos x \, dx$ is

a) $-\frac{1}{4}$

b) $\frac{1}{4}$

c) $\frac{\pi}{2}$

d) $\frac{\pi}{4}$

ANSWER: (B)

$$\int f(x)f'(x)dx = \frac{[f(x)]^2}{2}$$

$f(x) = \sin x$, $f'(x) = \cos x$

$$= \left[\frac{\sin^2 x}{2}\right]_0^{\frac{\pi}{4}} = \frac{\sin^2\left(\frac{\pi}{4}\right)}{2} - 0$$

$$= \frac{\left(\frac{1}{\sqrt{2}}\right)^2}{2} = \frac{\frac{1}{2}}{2} = \frac{1}{4}$$

02

CHAPTER TWO

Statistics & Analytics

20 Marks

Questions 21 – 40

Q.21

DATA

The data extracted from the Google forms has the file extension as

a) .doc

b) .docx

c) .csv

d) .xlsx

✓ **ANSWER: (C)**

Data from Google Forms is exported as `.csv` (Comma-Separated Values).

Q.22

DATA CLASSIFICATION

Match the following:

COLUMN A	COLUMN B
a) Data	i) Survey
b) Questionnaire	ii) Number of students
c) Interview	iii) Google form
d) Quantitative data	iv) Facts and figure

a) a-iv, b-i, c-iii, d-ii

b) a-iv, b-ii, c-iii, d-i

c) a-iv, b-iii, c-i, d-ii

d) a-iv, b-i, c-ii, d-iii

✓ **ANSWER: (C)**

Data = facts & figures (iv), Questionnaire = Google form (iii), Interview = method of survey (i), Quantitative = numerically measurable like number of students (ii).

Q.23

DATA CLEANING

Data cleaning refers to

a) Modifying the gathered data

b) Adding data to the gathered data

c) Removing the data from gathered data

d) Processing the data according to the defined problem statement

✓ ANSWER: (A/B/C/D)

All options are objectives of **data cleaning**: modifying, adding, removing, and processing data.

Q.24

EXCEL

The syntax to find frequency distribution in an interval in Excel is

a) =COUNTIFS(criteria_range1, criteria1, ...)

b) =COUNTIF(criteria_range1, criteria1, ...)

c) =COUNT(criteria_range1, criteria1, ...)

d) =COUNTIFS(criteria1, criteria2, ...)

✓ ANSWER: (A)

The syntax is `=COUNTIFS(criteria_range1, criteria1, .....)`

Q.25

EXCEL

Which tab in Excel supports in plotting Pie chart?

a) Home tab

b) View tab

c) Insert tab

d) Data tab

✓ ANSWER: (C)

The **Insert tab** is used to plot pie charts in Excel.

Q.26

CHARTS

Frequency polygon is a __________ graph

a) Scatter graph

b) Pie graph

c) Bar graph

d) Line graph

✓ ANSWER: (D)

A frequency polygon is a **line graph**.

Q.27

HISTOGRAM

Which of the following statements on Histogram is true?

a) A Histogram can display categorical data effectively

b) The bars in a Histogram should not touch each other

c) A Histogram shows the frequency distribution of continuous data

d) A Histogram is same as a bar chart

✓ ANSWER: (C)

A histogram shows **frequency distribution of continuous data**.

Q.28

STEM-LEAF

Choose the right stem-leaf plot for data: 52, 20, 44, 28, 20, 41, 30, 56, 32, 28

Plot-1:

STEM	LEAF
2	0088
3	20
4	14
5	26

Plot-2:

STEM	LEAF
2	0088
3	02
4	14
5	26

Plot-3:

STEM	LEAF
2	08
3	20
4	14
5	26

Plot-4:

STEM	LEAF
2	80
3	20
4	41
5	62

a) Plot-1

b) Plot-2

c) Plot-3

d) Plot-4

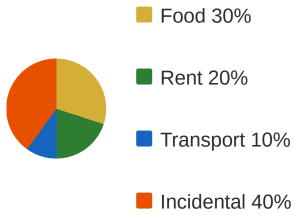
✓ ANSWER: (B)

Arranging data: 20,20,28,28,30,32,41,44,52,56. Stem|Leaf: 2|0088, 3|02, 4|14, 5|26. Key: 2|0=20.

Q.29

PIE CHART

From a pie chart: Food 30%, Rent 20%, Transport 10%, Incidental 40%. Salary = ₹50,000. Total expenditure on rent and transport is?



a) ₹30,000

b) ₹25,000

c) ₹15,000

d) ₹10,000

✓ ANSWER: (C)

Rent + Transport = 20% + 10% = 30%

$= \frac{30}{100} \times 50000 = ₹15,000$

Q.30

STATISTICS

For data: 28, 36, 44, 58, 44, 33, 76, 59, 58, 39, 58. Match the following:

MEASURE	VALUE
i) Range	a) 58
ii) Mean	b) 44
iii) Mode	c) 48.4545
iv) Median	d) 48

a) i-d, ii-c, iii-b, iv-a

b) i-d, ii-c, iii-a, iv-b

c) i-a, ii-b, iii-c, iv-d

d) i-d, ii-a, iii-b, iv-c

✓ ANSWER: (B)

Range = 76−28 = 48 (d)

Mean $= \frac{533}{11} = 48.4545$ (c)

Mode = 58 (most frequent) (a)

Median = 44 (middle value) (b)

Q.31

EXCEL

The Excel syntax for finding quartile 4 (4th quartile) for a given data set is

a) =QUARTILE [array, 4]

b) =QUARTILE (array, 4)

c) =QUARTILE (array_4)

d) =QUARTILE (array-4)

✓ ANSWER: (B)

The syntax is `=QUARTILE(array, 4)`

Q.32

STATISTICS

Which of the following equalities and inequalities is true for the data set 25, 25, 25, 25, 25, 25, 25, 25?

a) Variance > Standard deviation

b) Variance = Standard deviation

c) Variance < Standard deviation

d) Variance < 0, Standard deviation < 0

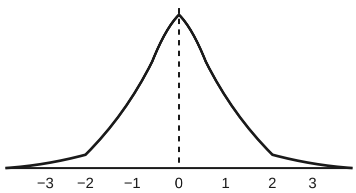
✓ ANSWER: (B)

All values are identical, so Variance = SD = 0. Hence **Variance = Standard deviation**.

Q.33

DISTRIBUTION

Skewness of the symmetric distribution shown is



a) > 0

b) = 0

c) < 0

d) = 1

✓ ANSWER: (B)

In a symmetric distribution, skewness β_1 = **0**.

Q.34

KURTOSIS

If the Kurtosis = 3, then the data distribution is called

a) Leptokurtic

b) Mesokurtic

c) Platykurtic

d) Skewkurtic

✓ ANSWER: (B)

When $\beta_2 = 3$, the distribution is **mesokurtic** (normal distribution).

Q.35

PYTHON

Assertion (A): An "int" type data in Python can store floating point numbers.
Reasoning (R): Python "int" type only stores positive integers.

a) A is true, R is false

b) A is false, R is true

c) A is false, R is false

d) A is true, R is true

✓ ANSWER: (C)

A is false — `int` cannot store floats. R is also false — `int` stores positive, negative, and zero.

Q.36

PYTHON

Match the data types:

EXPRESSION	DATA TYPE
a) a = 5	i) String
b) a or b	ii) Float
c) "Hello KEA"	iii) Integer
d) C = 5.9	iv) Boolean

a) a-iii, b-iv, c-i, d-ii

b) a-iv, b-i, c-ii, d-iii

c) a-iv, b-i, c-iii, d-ii

d) a-i, b-ii, c-iii, d-iv

✓ ANSWER: (A)

a=5 → Integer(iii), a or b → Boolean(iv), "Hello KEA" → String(i), C=5.9 → Float(ii)

Q.37

PYTHON

What is the syntax to assign the data type to a variable in Python?

a) Name = "hello world"

b) Height = 5.9

c) Assertion = True

d) All of the above

✓ ANSWER: (D)

Python uses **dynamic typing** — data types are assigned automatically based on the value.

Q.38

PYTHON

Which of the following is a valid variable name in Python?

a) 1 Var

b) Var_name

c) Var - name

d) Var name

✓ ANSWER: (B)

Var_name is valid — starts with letter, contains only letters and underscore, no spaces or special characters.

Q.39

PYTHON

Output of the following Python program is:

```
n = 5
A = 0
while (n > 0) :
    A = A + n
    n = n - 1
print (A)
```

a) 15

b) 5

c) 55

d) 0

✓ ANSWER: (A)

```
n=5: A=0+5=5, n=4: A=5+4=9,
n=3: A=9+3=12, n=2: A=12+2=14,
n=1: A=14+1=15, n=0: loop stops.
print(A) → 15
```

Q.40

PYTHON

Which of the following is the correct syntax of an if statement in Python?

a) if x>0 then:

b) if (x>10)

c) if x>10:

d) if x>10;

✓ ANSWER: (C)

Python if syntax: `if condition:` — no parentheses needed, must end with colon.

03

CHAPTER THREE

IT Skills

20 Marks

Questions 41 – 60

Q.41

ALGORITHMS

Consider: Step 1: A=5, B=3. Step 2: if A>B, then A=A−B. Step 3: B=A+B. Step 4: Output B. What is the output?

a) 2

b) 5

c) 8

d) 3

✓ ANSWER: (B)

```
A=5, B=3. A>B? Yes → A=5−3=2
B = A+B = 2+3 = 5. Output B = 5
```

Q.42

ALGORITHMS

Which of the following best defines an algorithm?

a) A graphical representation

b) A code snippet

c) A step-by-step procedure to solve a problem

d) A mobile application

✓ ANSWER: (C)

An algorithm is a **finite sequence of well-defined steps** to solve a problem.

Q.43

ALGORITHMS

Consider the following algorithm:

```
Start
if a <= b and a <= c
    print a
else if b <= a and b <= c
    print b
else
    print c
End.
```

if a = 10, b = 25, c = 33

What output will be printed for the above algorithm?

a) 25

b) 10

c) 33

d) None of the above

✓ ANSWER: (B)

10<=25 (True) and 10<=33 (True) → both conditions true → prints **10**.

Q.44

ALGORITHMS

Consider the following algorithm:

```
if num > 0
    num = num + 10
    print num
else if num == 0
    num = num * 10
    print num
else
    print num.
```

What will be the output, if num = −5?

a) 50

b) −50

c) −5

d) 5

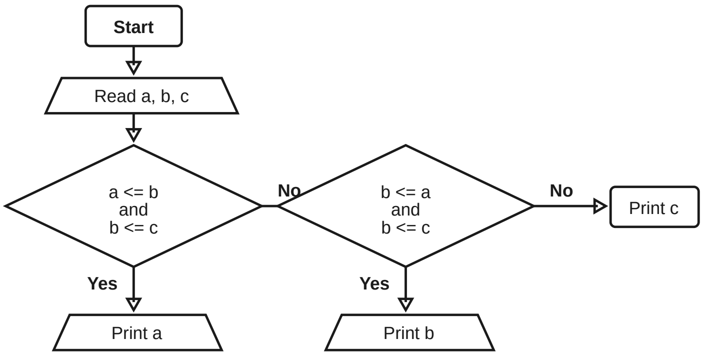
✓ ANSWER: (C)

−5>0? No. −5==0? No. else: print num → prints **−5**.

Q.45

FLOWCHARTS

Consider the following flowchart.



What will be the output if a = 10, b = 15, c = 5?

a) 10

b) 15

c) error

d) 5

✓ ANSWER: (D)

a<=b and b<=c → 10<=15 (T) and 15<=5 (F) → False. b<=a and b<=c → 15<=10 (F) → False. else → print c = **5**.

Q.46

WEB

Which of the following statements about web browsers is correct?

a) Web browsers translate HTML, CSS and JavaScript into machine code before displaying a web page to the user.

b) Web browsers can only display content written in HTML and CSS, but they cannot execute JavaScript.

c) Web browsers interpret HTML, CSS & JavaScript to render a webpage allowing dynamic content and interaction.

d) Web browsers use server-side programming languages like PHP to render webpages directly

✓ ANSWER: (C)

Browsers **interpret HTML, CSS & JavaScript** to render webpages.

Q.47 HTML

Consider the following HTML code snippet.

```
<select name = "cars" id = "carselect">
  <option value = "volvo">Volvo</option>
  <option value = "XUV">XUV</option>
  <option value = "Tata">Tata</option>
</select>
```

What does the <select> element do in this code?

a) It creates a text input field for users to type in

b) It creates a dropdown list from which the user can select a car brand.

c) It allows users to choose multiple options from a list of cars.

d) It creates a button that users can click to select a car brand

✓ **ANSWER: (B)**

The `<select>` element creates a **dropdown list**.

Q.48 CSS

Which CSS property is used to set the background color of a webpage?

a) background-color

b) color

c) bg-color

d) background-image

✓ **ANSWER: (A)**

`background-color` sets the background color of an element.

Q.49 CSS

Which CSS selector is used to select an HTML element with a specific id?

a) .id

b) #id

c) id

d) [id]

✓ **ANSWER: (B)**

The hash symbol `#` is the ID selector in CSS.

Q.50

JAVASCRIPT

Which event is triggered when a user clicks on an HTML element in JavaScript?

a) onload

b) onclick

c) onhover

d) onchange

✓ ANSWER: (B)

The `onclick` event fires when a user clicks on an element.

Q.51

SEO

Which factor can help improve a website's SEO ranking?

a) Using JavaScript for content rendering

b) High Keyword density in the content

c) Using relevant and high-quality backlinks

d) Including as many images as possible

✓ ANSWER: (C)

Backlinks from reputable websites are "votes of confidence" for SEO.

Q.52

ORGANIZATION

Which of the following are the correct description with respect to structured organisation.
(a) Operations → produces goods or services, ensures quality and efficiency.
(b) Marketing → Identifies customer needs, promotes or services.
(c) H.R. (Human Resources): Manages budgeting, accounting and financial planning.

a) (a), (b)

b) (a), (b), (c)

c) (a) only

d) (b) only

✓ ANSWER: (A)

HR does not handle budgeting/financial planning — that's Finance. Only **(a) Operations and (b) Marketing** are correct.

Q.53
ERP

Which of the following statements best describes the purpose of ERP tools in an organisation?

a) ERP tools are designed to manage customer relationships and marketing strategies.

b) ERP tools are used to integrate and automate key business processes across department as finance, HR, inventory and supply chain.

c) ERP tools are primarily focused in managing email communication between employees an organization.

d) ERP tools are closely used for managing the IT infrastructure and hardware resources organisation.

✓ **ANSWER: (B)**

ERP systems **unify operations across departments** for seamless coordination.

Q.54
CLOUD

In cloud computing, data is stored in

a) Local folders

b) Physical notebooks

c) Remote servers accessed via the internet

d) USB drives

✓ **ANSWER: (C)**

Cloud computing stores data on **remote internet-connected servers**.

Q.55
IOT

Which of the following is an example of IoT in smart homes?

a) A GPS system in a car for navigation

b) A thermostat that adjusts temperature based on occupancy and preferences

c) An e-commerce website for shopping

d) A manual doorbell

✓ **ANSWER: (B)**

Smart **thermostats** use sensors and connectivity — a core IoT feature in smart homes.

Q.56

CLOUD

Which of the following is an example of Software-as-a-Service (SaaS)?

a) Microsoft Azure

b) Google Drive

c) Amazon Web Services (AWS) EC2

d) Google Kubernetes Engine

✓ ANSWER: (A & B)

SaaS delivers applications over the internet. **Microsoft Azure** also provides SaaS, and **Google Drive** is a SaaS application.

Q.57

IOT

Which of the following is a platform for developing IoT applications?

a) Google Docs

b) Arduino

c) Microsoft

d) WinZip

✓ ANSWER: (B)

Arduino is an open-source electronics platform for building IoT devices.

Q.58

CYBER SECURITY

Ravi often uses incognito mode while browsing, believing it protects him from all types of tracking and threats. He does not use antivirus software or phishing protection.
Which of the following statement about Ravi's assumption is correct?

a) Incognito mode provides full malware protection

b) Incognito mode hides activity from the device and protects against all trackers

c) Incognito mode only prevents local history tracking not online threats

d) Incognito mode encrypts his traffic and login credentials

✓ ANSWER: (C)

Incognito mode only stops saving local history, cookies, and site data. It does **not protect** from malware or phishing.

Q.59

CYBER SAFETY

Rita uses a public computer at a cyber cafe for email and closes the browser without logging out. What cyber safety mistake did she make?

a) Read personal emails in public

b) Used a public computer

c) She didn't logout of her account

d) She didn't clear browsing history

✓ ANSWER: (C)

Not logging out on public computers poses a risk of unauthorized access.

Q.60

CYBER SECURITY

To safeguard against cyber attacks that steal sensitive information, which browser setting should be enabled?

a) Enable automatic updates

b) Turn on phishing and malware protection

c) Disable phishing and malware protection

d) Enable cookies

✓ ANSWER: (B)

Phishing and malware protection warns users about harmful websites and downloads.

04

CHAPTER FOUR

Fundamentals of Electrical & Electronics Engineering

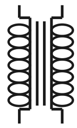
20 Marks

Questions 61 – 80

Q.61

SYMBOLS

Match the following with symbols

COMPONENT	SYMBOL
i) Resistance	a) 
ii) Transformer	b) 
iii) Capacitor	c) 
iv) Battery	d) 

a) i-a, ii-c, iii-d, iv-b

b) i-d, ii-b, iii-c, iv-a

c) i-b, ii-c, iii-d, iv-a

d) i-d, ii-c, iii-a, iv-b

✓ **ANSWER: (D)**

Resistance → zigzag, Transformer → coils, Capacitor → parallel lines, Battery → long/short lines.

Q.62

EARTHING

Earthing plate is made up of

a) Bakelite

b) Aluminum

c) Copper

d) Plastic

✓ **ANSWER: (C)**

Copper is a good conductor and corrosion-resistant, ideal for earthing.

Q.63

UNITS

Match the following:

QUANTITY	UNIT
i) Current	a) Watts
ii) Power	b) Hertz
iii) Energy	c) kWh
iv) Frequency	d) Ampere

a) i-a, ii-b, iii-d, iv-c

b) i-a, ii-c, iii-b, iv-d

c) i-c, ii-b, iii-d, iv-a

d) i-d, ii-a, iii-c, iv-b

✓ ANSWER: (D)

Current → Ampere, Power → Watts, Energy → kWh, Frequency → Hertz.

Q.64

CIRCUITS

If 12Ω and 6Ω are connected in parallel, the equivalent resistance is

a) 18 Ω

b) 06 Ω

c) 04 Ω

d) 02 Ω

✓ ANSWER: (C)

$$\frac{1}{R} = \frac{1}{12} + \frac{1}{6} = \frac{1+2}{12} = \frac{3}{12} = \frac{1}{4}$$

$$R = 4\ \Omega$$

Q.65

OHM'S LAW

In a circuit with R=6Ω and V=12V, the current is

a) 72A

b) 0.5A

c) 2A

d) 1.2A

✓ ANSWER: (C)

$$I = \frac{V}{R} = \frac{12}{6} = 2A$$

Q.66

AC THEORY

The frequency of a signal with period 0.02 sec is

a) 100 Hz

b) 200 Hz

c) 50 Hz

d) 150 Hz

✓ ANSWER: (C)

$$f = \frac{1}{T} = \frac{1}{0.02} = 50\ \text{Hz}$$

Q.67

AC THEORY

In a pure resistive circuit, which statement is true?

a) Both current and voltage are in phase

b) Current leads the voltage

c) Voltage lags the current

d) Both are out of phase

✓ ANSWER: (A)

In a pure resistive AC circuit, current and voltage are **in phase**.

Q.68

PROTECTION

HRC fuses can be used up to a maximum current of

a) 250 mA

b) 500 mA

c) 1 A

d) 1250 A

✓ ANSWER: (D)

HRC (High Rupturing Capacity) fuses handle very high fault currents, up to **1250 A or more**.

Q.69

WIRING

To control a single lamp from two different places, the switch used is

a) Double pole

b) Two way

c) Single pole

d) One way

✓ ANSWER: (B)

Two-way switches allow control from two different locations.

Q.70

FUSE

A fuse is always connected ___________ to the circuit.

a) Parallel

b) Across the load

c) Series

d) Across the supply

ANSWER: (C)

A fuse is connected in **series** to break the circuit on overload.

Q.71

TRANSFORMER

In a Transformer, ___________ remains constant

a) Voltage

b) Frequency

c) Current

d) Both current and voltage

ANSWER: (B)

In both step-up and step-down transformers, **frequency** remains the same (e.g., 50 Hz).

Q.72

BATTERIES

The combination of two or more cells is

a) Battery

b) load cell

c) Primary cell

d) Generator

ANSWER: (A)

A **battery** is formed by connecting two or more cells.

Q.73

MATCHING

Match the following:

	COLUMN A		COLUMN B
i)	Ampere hour	a)	1φ Induction motor
ii)	KVA	b)	Lead acid battery
iii)	Star delta	c)	Transformer
iv)	Fan	d)	3φ Induction motor

a) i-a, ii-b, iii-c, iv-d

b) i-c, ii-a, iii-d, iv-b

c) i-b, ii-c, iii-d, iv-a

d) i-b, ii-a, iii-c, iv-d

✓ **ANSWER: (C)**

Ampere-hour → Battery, KVA → Transformer, Star-delta → 3φ motor, Fan → 1φ motor.

Q.74

TRANSFORMER

In a transformer, the winding to which the load is connected is called __________ winding

a) Primary

b) Axillary

c) Secondary

d) Concentric

✓ **ANSWER: (C)**

The **secondary winding** is connected to the load and delivers output voltage.

Q.75

TRANSFORMER

Which statement is FALSE with respect to transformer?

a) Converts high voltage to low voltage

b) Converts high frequency to low frequency

c) Converts low voltage to high voltage

d) Transfers energy at constant frequency

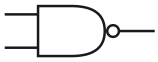
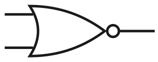
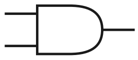
✓ **ANSWER: (B)**

A transformer does **not change frequency**. It can step up or step down voltage, but frequency remains unchanged.

Q.76

LOGIC GATES

Match the following logic gates with their symbols:

GATE	SYMBOL
i) AND	a) 
ii) OR	b) 
iii) NAND	c) 
iv) NOR	d) 

a) i-a, ii-b, iii-c, iv-d

b) i-b, ii-c, iii-d, iv-a

c) i-d, ii-a, iii-c, iv-b

d) i-c, ii-d, iii-a, iv-b

✓ ANSWER: (D)

AND, OR, NAND, NOR matched with their respective standard symbols.

Q.77

SENSORS

Which is a common type of sensor used for measuring temperature?

a) Proximity sensor

b) Pressure sensor

c) Flow sensor

d) Thermocouple

✓ ANSWER: (D)

A **thermocouple** converts thermal energy into electrical voltage for temperature measurement.

Q.78

RESISTORS

What does the fourth band in a 4-band resistor indicate?

a) Resistance

b) Multiplier

c) Power rating

d) Tolerance

✓ ANSWER: (D)

1st & 2nd bands = significant digits, 3rd = multiplier, 4th = **tolerance**.

Q.79

MATERIALS

Which of the following is an example of Insulator?

a) Aluminium

b) Rubber

c) Iron

d) Silicon

✓ ANSWER: (B)

Rubber resists the flow of electric current. Aluminium and Iron are conductors; Silicon is a semiconductor.

Q.80

ELECTRONICS

Diode is made up of __________ material.

a) Conducting

b) Semiconducting

c) Insulating

d) Electronic

✓ ANSWER: (B)

Diodes are made from **semiconductor** materials like Silicon or Germanium.

05

CHAPTER FIVE

Project Management Skills

20 Marks

Questions 81 – 100

Q.81

ANALOGY

Complete the analogy: Project : Deliverables :: Operation : __________

a) Deadlines
b) Efficiency
c) Repetition
d) Conclusion

✓ ANSWER: (C)

Projects have specific deliverables; operations are permanent processes with **repetitive** outcomes.

Q.82

OPERATIONS

Multiple objectives are achieved again and again in __________

a) Operation
b) Project
c) Business
d) Sales

✓ ANSWER: (A)

Operations are repetitive processes where objectives are achieved again and again.

Q.83

CONSULTANCY

Which are examples of consultancy firms? A) Accenture, B) TCS, C) Tata Motors, D) Birla Technical Services

a) A, B and C only
b) A, B and D only
c) A and D only
d) A, B, C and D

✓ ANSWER: (B)

Accenture, TCS, and Birla Technical Services are consultancy firms. **Tata Motors** is automotive manufacturing.

Q.84

PM BASICS

Which of the following is NOT an obstacle in project management?

a) Undefined project scope

b) Poor communication among team members

c) Clear project objectives and goals

d) Limited resources

✓ ANSWER: (C)

Clear project objectives is an advantage, not an obstacle.

Q.85

ANALOGY

Project Procedure Manual : Standard Operating Procedures :: Project Diary : ___________

a) Records daily events and issues

b) Provides step-by-step instructions

c) Outlines rules and processes

d) Used only during project closure

✓ ANSWER: (A)

A project diary **records daily events, activities, and issues** throughout the project life cycle.

Q.86

TEAMS

Match the following:

TEAM TYPE	ROLE
i) Initial project team	a) Guides the project with advice and support
ii) Core project team	b) Works on the project every day, making decisions and doing tasks
iii) Project stakeholders	c) People interested in the project outcomes like clients or sponsors
iv) Project advisors	d) Helps to start the project and plan the initial steps

a) i-d, ii-b, iii-c, iv-a

b) i-a, ii-b, iii-c, iv-d

c) i-b, ii-a, iii-c, iv-d

d) i-d, ii-c, iii-b, iv-a

✓ ANSWER: (A)

Initial team starts the project (d), Core team works daily (b), Stakeholders are interested parties (c), Advisors guide with advice (a).

Q.87

WBS

What does Work Breakdown Structure (WBS) do to help a Project Manager?

a) Track the project's financial performance

b) Breakdown the project into smaller, manageable tasks

c) Determine project's risk level

d) Allocate resources to team members

✓ ANSWER: (B)

WBS **breaks down complex work into smaller, manageable tasks**.

Q.88

PEP

Which is NOT a subpart of Project Execution Plan (PEP)?

a) Organizational plan

b) Work packing plan

c) Staffing plan

d) Contracting plan

✓ ANSWER: (C)

Staffing plan is not included in PEP.

Q.89

SCENARIOS

High demand, high selling price, low variable cost are examples of

a) Normal Scenario

b) Worst Scenario

c) Best Scenario

d) Abnormal Scenario

✓ ANSWER: (C)

High demand, high selling price, low cost represent the **best scenario**.

Q.90

LIFE CYCLE

In __________ phase, project definition and problem identification are defined

a) Planning

b) Execution

c) Initiation

d) Closure

✓ ANSWER: (C)

Initiation is the first phase where project is defined and problems are identified.

Q.91

RISK

Risk arising due to fluctuation of interest rate OR mistakes in accounting is known as __________ risks.

a) Policy

b) Technical

c) Marketing

d) Financial

✓ ANSWER: (D)

Financial risks arise from interest rate fluctuations or accounting errors.

Q.92

ASSERTION

Assertion (A): Time overrun can lead to cost overrun in a project.
Reason (R): More time often means more money spent on labour and resources.

a) Both (A) and (R) are true and (R) is the correct explanation of (A)

b) Both (A) and (R) are true but (R) is not the correct explanation of (A)

c) (A) is true, but (R) is false

d) (A) is false, but (R) is true

✓ ANSWER: (A)

Both statements are true, and the reason correctly explains why time overrun leads to cost overrun.

Q.93

CPM

The full form of CPM in project management is

a) Critical Process Management

b) Critical Path Method

c) Continuous Project Monitoring

d) Cost Planning Model

✓ ANSWER: (B)

CPM stands for **Critical Path Method**.

Q.94

TOOLS

Which is NOT a tool of project planning, scheduling and monitoring?

a) Gantt Chart

b) Flow chart

c) PERT Chart

d) Milestone Chart

✓ ANSWER: (B)

Flow chart is not a project planning/scheduling/monitoring tool.

Q.95

EVALUATION

Project Evaluation can be done ___________

a) At the end of project only

b) Throughout the project lifecycle

c) Only during planning phase

d) At the start only

✓ ANSWER: (B)

Project evaluation is done **throughout the project lifecycle**.

Q.96

PLANNING

Identify the incorrect statement regarding project planning functions:

a) Defines project goals and objectives

b) Allocates resources without considering constraints

c) Establishes a project timeline and milestones

d) Identifies potential risks and mitigates them

✓ ANSWER: (B)

Project planning allocates resources **by considering constraints**, not without.

Q.97

SMART

Which is NOT a key component of "SMART" goal?

a) Time-Bound

b) Random

c) Achievable

d) Specific

✓ ANSWER: (B)

SMART = Specific, Measurable, Achievable, Realistic, Timely. **Random** is not a component.

Q.98

REVIEW

The process of correcting and rectifying the errors in the project is

a) Project Audit

b) Project Monitoring

c) Project Evaluation

d) Project Review

ANSWER: (D)

Project Review is the process of correcting and rectifying errors.

Q.99

PERT

Optimistic Time = 4 days, Pessimistic Time = 8 days, Most likely time = 6 days. The expected time to complete is

a) 7 days

b) 6 days

c) 5 days

d) 4 days

ANSWER: (B)

$$T_e = \frac{t_o + 4t_m + t_p}{6}$$

$$= \frac{4 + 4(6) + 8}{6} = \frac{36}{6} = 6 \text{ days}$$

Q.100

TECHNOLOGY

Watching 3D movie is an example of

a) Augmented Reality

b) Virtual Reality

c) Augmented and Virtual Reality

d) Cloud Technology

ANSWER: (A, B, C)

Watching a 3D movie is an example of **Augmented Reality** , **Virtual Reality** , and both combined.

2026

PART FOUR

DCET 2026

100 Questions • 5 Subjects • 100 Marks

Previous Year Question Paper & Complete Solutions

01

CHAPTER 1

Mathematics

20 Questions · 20 Marks

DCET 2026 — Set D1

Q.01

MATH

If A is a square matrix given by $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$, the principal diagonal elements of above matrix are

(1) 1 2 3

(2) 4 5 6

(3) 1 5 9

(4) 7 5 3

✓ ANSWER: (3)

The principal diagonal of a matrix runs from top-left to bottom-right. For the given 3×3 matrix, the diagonal elements are $\boxed{a_{11} = 1, a_{22} = 5, a_{33} = 9}$.

Q.02

MATH

The value of $\begin{vmatrix} \text{Cosx} & -\text{Sinx} & 0 \\ \text{Sinx} & \text{Cosx} & 0 \\ 0 & 0 & 1 \end{vmatrix}$ is

(1) −1

(2) 0

(3) 1

(4) Cos2x

✓ ANSWER: (3)

> Expanding along R_3: $1 \times (\text{Cos}^2x + \text{Sin}^2x) = 1$

Using the identity $\text{Cos}^2x + \text{Sin}^2x = 1$, the determinant evaluates to $\boxed{1}$.

Q.03

MATH

For any square matrix A, identify the correct statement

(1) $A \cdot \text{adj } A = |A|I$

(2) $A \cdot \text{adj } A = 0$

(3) $\text{adj } A = |A|I$

(4) $A \cdot \text{adj } A = 1$

✓ ANSWER: (1)

The standard identity for any square matrix: `A · adj(A) = |A| · I`, where adj(A) is the adjugate matrix and I is the identity matrix.

Q.04

MATH

The minors of the elements 5 & 2 in the matrix $A = \begin{bmatrix} 1 & -2 & 4 \\ 5 & 2 & -1 \\ 7 & 6 & 9 \end{bmatrix}$ are

(1) 24, 52

(2) −42, −19

(3) −6, 2

(4) 42, 19

✓ ANSWER: (2)

$$\text{Minor of } 5 = \begin{vmatrix} -2 & 4 \\ 6 & 9 \end{vmatrix} = (-2)(9) - (4)(6) = -18 - 24 = -42$$

$$\text{Minor of } 2 = \begin{vmatrix} 1 & 4 \\ 7 & 9 \end{vmatrix} = (1)(9) - (4)(7) = 9 - 28 = -19$$

Q.05

MATH

If $x + y + \lambda = 0$ is an equation of a line passing through the point (1, 2), then the value of λ is

____

(1) 3

(2) $\frac{1}{3}$

(3) −3

(4) $-\frac{1}{3}$

✓ ANSWER: (3)

Substituting (1, 2): $1 + 2 + \lambda = 0$

$3 + \lambda = 0 \rightarrow \lambda = -3$

Q.06

MATH

If $4y - 3x - 7 = 0$ is a given equation of a straight line, then its slope is ___________

(1) $\frac{4}{3}$

(2) $-\frac{4}{3}$

(3) $\frac{3}{4}$

(4) $-\frac{3}{4}$

✓ ANSWER: (3)

$$4y - 3x - 7 = 0$$
$$4y = 3x + 7$$
$$y = \left(\frac{3}{4}\right)x + \frac{7}{4}$$
$$\text{Slope } m = \frac{3}{4}$$

Q.07

MATH

If a straight line is parallel to Y-axis, then its slope is ___________

(1) 0

(2) ∞

(3) 1

(4) -1

✓ ANSWER: (2)

A line parallel to the Y-axis is vertical. The slope of a vertical line is **undefined (∞)**, as the change in x is zero.

Q.08

MATH

Equation of a straight line whose angle of inclination with x-axis is 135° and having 4 units as its y-intercept is ___________

(1) $x - y - 4 = 0$

(2) $x - y + 4 = 0$

(3) $x + y + 4 = 0$

(4) $x + y - 4 = 0$

✓ ANSWER: (4)

$$\text{Slope } m = \tan(135°) = -1$$
$$\text{y-intercept } c = 4$$
$$y = -x + 4 \rightarrow x + y - 4 = 0$$

Q.09

MATH

If $y = \frac{2}{(1-x)}$, then dy/dx is

(1) 0

(2) $\frac{2}{(1-x)^2}$

(3) $\frac{1}{(1-x)^2}$

(4) $-\frac{2}{(1-x)^2}$

✓ ANSWER: (2)

$y = 2(1-x)^{-1}$

$dy/dx = 2 \times (-1)(1-x)^{-2} \times (-1)$

$dy/dx = \frac{2}{(1-x)^2}$

Q.10

MATH

The derivative of $e^x \cos x$ is

(1) $e^x(\sin x - \cos x)$

(2) $e^x(-\sin x - \cos x)$

(3) $e^x(-\sin x + \cos x)$

(4) $-e^x(-\sin x + \cos x)$

✓ ANSWER: (3)

$d/dx\,[e^x \cdot \cos x] = e^x \cdot (-\sin x) + \cos x \cdot e^x$

$= e^x(\cos x - \sin x)$

$= e^x(-\sin x + \cos x)$

Q.11

MATH

If $y = e^{3x} + e^{-5x}$ then d^2y/dx^2 at $x = 0$ is

(1) 35

(2) 34

(3) 36

(4) −35

✓ ANSWER: (2)

$dy/dx = 3e^{3x} - 5e^{-5x}$

$d^2y/dx^2 = 9e^{3x} + 25e^{-5x}$

At $x = 0$: $9(1) + 25(1) = 34$

Q.12

MATH

Rate of change of area 'A' of a circle with respect to its radius 'r' is

(1) $dA/dr = 2\pi r$

(2) $dA/dr = \pi r^2$

(3) $dA/dr = 2\pi$

(4) $dA/dr = \pi r$

✓ ANSWER: (1)

$A = \pi r^2$

$dA/dr = 2\pi r$

Q.13

MATH

Match List – I (function) with List – II (integrand) and select the correct answer:

LIST – I (FUNCTION)	LIST – II (INTEGRAND)
(a) $\int e^x\,dx$	(i) $\log x + c$
(b) $\int\left(\frac{1}{x}\right) dx$	(ii) $e^x + c$
(c) $\int dx$	(iii) $\tan x + c$
(d) $\int \sec^2 x\,dx$	(iv) $x + c$

(1) a – i, b – ii, c – iv, d – iii

(2) a – ii, b – iii, c – i, d – iv

(3) a – ii, b – i, c – iv, d – iii

(4) a – iv, b – iii, c – ii, d – i

✓ ANSWER: (3)

$\int e^x\,dx = e^x + c \rightarrow$ a-ii; $\int\left(\frac{1}{x}\right) dx = \log x + c \rightarrow$ b-i; $\int dx = x + c \rightarrow$ c-iv; $\int \sec^2 x\,dx = \tan x + c \rightarrow$ d-iii.

Q.14

MATH

$\int \sin^6 x \cdot \cos x \, dx$

(1) $\frac{\cos^7 x}{7} + c$

(2) $-\frac{\cos^7 x}{7} + c$

(3) $\frac{\sin^7 x}{7} + c$

(4) $6 \sin^6 x + c$

✓ ANSWER: (3)

Let $u = \sin x$, $du = \cos x \, dx$

$\int u^6 \, du = \frac{u^7}{7} + c = \frac{\sin^7 x}{7} + c$

Q.15

MATH

The integral of x cosx with respect to 'x' is

(1) x cosx + sinx + c

(2) x sinx + cosx + c

(3) cosx + sinx + c

(4) x(sinx + cosx) + c

✓ ANSWER: (2)

Using integration by parts: $\int x \cdot \cos x \, dx$

$u = x$, $dv = \cos x \, dx$

$= x \cdot \sin x - \int \sin x \, dx$

$= x \sin x + \cos x + c$

Q.16 MATH

The volume of solid generated by revolving the curve $y = \sqrt{x^2 + 5x}$ between $x = 1$ and $x = 2$ is

(1) $\frac{59\pi}{6}$ cubic units

(2) $\frac{59}{6}$ cubic units

(3) $\frac{58\pi}{5}$ cubic units

(4) $\frac{59\pi}{5}$ cubic units

✓ ANSWER: (1)

$$V = \pi\int_1^2 y^2\,dx = \pi\int_1^2 (x^2 + 5x)\,dx$$
$$= \pi\left[\frac{x^3}{3} + \frac{5x^2}{2}\right]_1^2$$
$$= \pi\left[\left(\frac{8}{3} + 10\right) - \left(\frac{1}{3} + \frac{5}{2}\right)\right]$$
$$= \pi\left[\frac{38}{3} - \frac{11}{6}\right] = \pi \times \frac{59}{6} = \frac{59\pi}{6}$$

Q.17 MATH

The value of $\sin 480°$ is

(1) $\frac{\sqrt{3}}{2}$

(2) $\sqrt{\left(\frac{2}{3}\right)}$

(3) $\sqrt{3}$

(4) $\frac{\sqrt{3}}{2} \times 2$

✓ ANSWER: (1)

$$\sin 480° = \sin(480° - 360°) = \sin 120°$$
$$= \sin(180° - 60°) = \sin 60° = \frac{\sqrt{3}}{2}$$

Q.18 MATH

Sin (A + B) is equal to

(1) SinA · CosB − CosA · SinB

(2) SinA · CosB + CosA · SinB

(3) CosA · CosB + SinA · SinB

(4) CosA · CosB − SinA · SinB

✓ ANSWER: (2)

The compound angle formula: `Sin(A + B) = SinA · CosB + CosA · SinB`.

Q.19

MATH

If $\cos A = \frac{12}{13}$ and $\sin B = \frac{3}{5}$, and A and B are acute angles, the value of cos(A − B) is

(1) $\frac{56}{65}$

(2) $-\frac{16}{65}$

(3) $\frac{33}{65}$

(4) $\frac{63}{65}$

✓ ANSWER: (4)

$\cos A = \frac{12}{13} \rightarrow \sin A = \frac{5}{13}$

$\sin B = \frac{3}{5} \rightarrow \cos B = \frac{4}{5}$

$\cos(A-B) = \cos A \cdot \cos B + \sin A \cdot \sin B$

$= \left(\frac{12}{13}\right)\left(\frac{4}{5}\right) + \left(\frac{5}{13}\right)\left(\frac{3}{5}\right)$

$= \frac{48}{65} + \frac{15}{65} = \frac{63}{65}$

Q.20

MATH

The value of sin 125° × sin 55° is equal to

(1) $\sin^2 35°$

(2) $\cos^2 35°$

(3) Sin 35°

(4) Cos 35°

✓ ANSWER: (2)

$\sin 125° = \sin(180° - 55°) = \sin 55° = \cos 35°$

$\therefore \sin 125° \times \sin 55° = \cos 35° \times \cos 35° = \cos^2 35°$

02

CHAPTER 2

Project Management

20 Questions · 20 Marks

DCET 2026 — Set D1

Q.21

PM

Complete the following Analogy:
Project : Temporary Endeavour :: Operation : ___________

(1) Ongoing work effort
(2) Plan
(3) Risk
(4) Deliverables

✓ **ANSWER: (1)**

A project is a **temporary endeavour**, whereas an operation is an **ongoing work effort** — it is continuous and repetitive, unlike the finite nature of a project.

Q.22

PM

Who provides guidance as well as direction to the projects?

(1) Manager
(2) Operator
(3) Consultants
(4) Administrator

✓ **ANSWER: (3)**

As per the answer key, **Consultants** provide guidance as well as direction to projects, offering expert advice on planning, execution and strategy.

Q.23

PM

Match the following project-based on their classification and select the correct option.

LIST – A	LIST – B
(a) Basis of sector	(i) Private sector project
(b) Basis of ownership	(ii) High risk project
(c) Basis of dependency	(iii) Industrial project
(d) Basis of risk involved in project	(iv) Dependent project

(1) a – iii, b – i, c – iv, d – ii

(2) a – iv, b – i, c – iii, d – ii

(3) a – iii, b – ii, c – iv, d – i

(4) a – ii, b – i, c – iv, d – iii

✓ ANSWER: (2)

Basis of sector → (iv), Basis of ownership → (i) Private sector, Basis of dependency → (iii) Industrial (dependent on industry), Basis of risk → (ii) High risk.

Q.24

PM

One of the misconceptions about project management is __________.

(1) Planning

(2) Rework

(3) Scheduling

(4) Controlling

✓ ANSWER: (2)

Rework is a common misconception — project management is about proactive planning, not redoing work. Planning, scheduling, and controlling are core PM functions.

Q.25

PM

Project work system can be designed by developing and preparing __________.

(1) Work breakdown system, Project Execution Plan, Project Procedure Manual

(2) Work breakdown system, Advance Action, Project Diary

(3) Work breakdown system, Project Execution Plan, Project Diary

(4) Work breakdown system, Project Execution Plan, Advance Action

✓ ANSWER: (1)

A project work system is designed using Work Breakdown System, Project Execution Plan, and Project Procedure Manual — these three form the backbone of project planning.

Q.26

PM

Choose the correct statements for successful project implementation:
(i) Adequate formulation
(ii) Sound project organisation
(iii) Proper implementation planning
(iv) Advance Action

(1) (i) and (ii) are correct

(2) (i), (ii) and (iii) are correct

(3) (ii) and (iii) are correct

(4) All are correct

✓ ANSWER: (4)

All four — adequate formulation, sound organisation, proper implementation planning, and advance action — are essential for successful project implementation.

Q.27

PM

In any project, who will maintain project diary?

(1) Financier

(2) Administrator

(3) Project Manager

(4) Accountant

✓ ANSWER: (3)

The **Project Manager** is responsible for maintaining the project diary, which records daily activities, progress, and decisions throughout the project lifecycle.

Q.28

PM

Match the following and choose the correct answer.

LIST – I	LIST – II
(a) Initial project team	(i) People who invested in the project
(b) Core project team	(ii) Group of people who give feedback and counselling
(c) Project Advisors	(iii) Specific people who initially conceive the idea
(d) Project stakeholders	(iv) Group ultimately responsible for designing and managing project

(1) a – ii, b – iii, c – i, d – iv

(2) a – ii, b – iii, c – iv, d – i

(3) a – iii, b – iv, c – ii, d – i

(4) a – i, b – iii, c – iv, d – ii

✓ ANSWER: (3)

Initial project team (a) → (iii) people who conceive the idea; Core project team (b) → (iv) responsible for designing/managing; Project Advisors (c) → (ii) give feedback; Stakeholders (d) → (i) investors.

Q.29

PM

Decision tree analysis in project management is used to evaluate __________ outcomes.

(1) Uncertainty

(2) Certainty

(3) Both (1) & (2)

(4) None of the above

✓ ANSWER: (3)

Decision tree analysis evaluates outcomes under both **uncertainty** and **certainty** conditions, helping project managers make informed decisions by mapping out possible scenarios.

Q.30

PM

What does the letter 'C' stand for in CPM?

(1) Crash

(2) Critical

(3) Cost

(4) Control

✓ ANSWER: (2)

CPM stands for **Critical Path Method**. The 'C' stands for Critical — a technique used to identify the longest path of activities in a project network.

Q.31

PM

In a network diagram, critical path is defined as __________

(1) Longest sequence of dependent activities with shortest project duration

(2) Longest sequence of dependent activities with longest project duration

(3) Shortest sequence of dependent activities with shortest project duration

(4) Shortest sequence of dependent activities with longest project duration

✓ ANSWER: (1)

The **critical path** is the longest sequence of dependent activities that determines the minimum (shortest) total project duration. Any delay on this path delays the entire project.

Q.32

PM

Match the project plan with the List.

LIST – I	LIST – II
(a) Resource plan	(i) Human risk
(b) Financial plan	(ii) Customer requirements
(c) Quality plan	(iii) Labour cost
(d) Risk plan	(iv) Type of labour

(1) a – i, b – ii, c – iii, d – iv

(2) a – ii, b – i, c – iv, d – iii

(3) a – iv, b – iii, c – ii, d – i

(4) a – iii, b – iv, c – i, d – ii

✓ ANSWER: (3)

Resource plan → (iv) Type of labour; Financial plan → (iii) Labour cost; Quality plan → (ii) Customer requirements; Risk plan → (i) Human risk.

Q.33

PM

Project planning involves __________

(1) Formal planning only

(2) Informal planning only

(3) Both (1) and (2)

(4) Neither (1) nor (2)

✓ ANSWER: (1)

Project planning primarily involves **formal planning** — documented plans including schedules, budgets, resource allocation, and risk management strategies.

Q.34

PM

Complex projects : Linear programming :: Large scale scheduling : __________

(1) Bar chart

(2) Network technique

(3) Gantt chart

(4) Heuristic programming

✓ ANSWER: (2)

Just as linear programming is used for complex projects, **Network technique** (like PERT/CPM) is used for large-scale scheduling to manage complex interdependencies.

Q.35

PM

Given List – I & List – II, match the following:

LIST – I	LIST – II
(a) Gantt chart	(i) Fix/set time target
(b) Network technique	(ii) Estimate time
(c) Project design	(iii) Interrelation activities
(d) Time Estimate	(iv) Blue print for action-oriented activity

(1) a – iii, b – ii, c – iv, d – i

(2) a – ii, b – iii, c – iv, d – i

(3) a – iii, b – ii, c – i, d – iv

(4) a – ii, b – iii, c – i, d – iv

✓ ANSWER: (2)

Gantt chart → (ii) Estimate time; Network technique → (iii) Interrelation activities; Project design → (iv) Blueprint for action; Time Estimate → (i) Fix/set time target.

Q.36

PM

Methodologies involved in planning is/are:
(i) Planning by incentive
(ii) Planning by direction

(1) (i) Only

(2) (ii) Only

(3) (i) and (ii)

(4) neither (i) nor (ii)

✓ ANSWER: (3)

Both **planning by incentive** and **planning by direction** are methodologies involved in project planning.

Q.37

PM

An audit of a project after it has been commissioned is known as ___________

(1) Performance evaluation

(2) Abandonment analysis

(3) Post-audit

(4) Initial review

✓ ANSWER: (3)

A **Post-audit** is an evaluation conducted after a project is completed and commissioned, to assess whether objectives were achieved and learn for future projects.

Q.38

PM

Which network technique is suitable for uncertain projects?

(1) Construction network

(2) PERT

(3) CPM

(4) Milestone chart

✓ ANSWER: (2)

PERT (Program Evaluation and Review Technique) is suitable for uncertain projects as it uses probabilistic time estimates (optimistic, most likely, pessimistic) to handle uncertainty.

Q.39

PM

Match the following and select the correct answer:

LIST – I	LIST – II
(a) IOT	(i) Gmail account
(b) Augmented reality	(ii) Smart farming
(c) Cloud technology	(iii) Gaming

(1) a – ii, b – iii, c – i

(2) a – ii, b – i, c – iii

(3) a – i, b – ii, c – iii

(4) a – iii, b – ii, c – i

✓ ANSWER: (1)

IoT → (ii) Smart farming (sensors in agriculture); Augmented reality → (iii) Gaming (AR overlays in games); Cloud technology → (i) Gmail account (cloud-based service).

Q.40

PM

The critical path in the given network diagram is __________
[Network: Start → A → B → C → E → End, A → C, A → D → E]

(1) A – B – E

(2) A – B – C – E

(3) A – D – E

(4) A – C – B – E

✓ ANSWER: (2)

The critical path A → B → C → E has the longest duration among all paths, making it the critical (longest) path that determines minimum project completion time.

03

CHAPTER 3

Applied Statistics & Data Analytics

20 Questions · 20 Marks

DCET 2026 — Set D1

Q.41

STATS

__________ is a method for graphically depicting groups of numerical data through their quartiles.

(1) Leaf – Plot

(2) Box – Plot

(3) Line – Plot

(4) Pie – Chart

✓ ANSWER: (2)

A **Box Plot** (box-and-whisker plot) displays data distribution using five-number summary: minimum, Q1, median, Q3, and maximum.

Q.42

STATS

Which of the following is true with respect to histogram?

(1) Histogram is represented for ordinal data set

(2) In histogram width is equal to the class interval and height is equal to relative frequency

(3) Histogram is used to represent continuous data which is equal to the width of class interval

(4) Histogram is used to represent the nominal data

✓ ANSWER: (3)

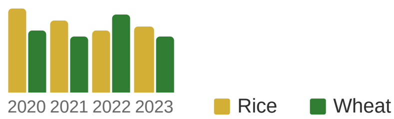
A **histogram** is used to represent continuous data where the width of each bar equals the class interval. It shows the frequency distribution of continuous variables.

Q.43

STATS

From the bar graph showing food grain production (Rice and Wheat) from 2020 to 2023:
In which years is rice production more than wheat?

Food Grain Production (in lakh tonnes)



(1) 2021 and 2022

(2) 2020 and 2023

(3) 2020 and 2021

(4) 2022 and 2023

✓ ANSWER: (3)

From the bar graph, rice production (hatched bars) exceeds wheat production (plain bars) in the years **2020 and 2021**.

Q.44

STATS

From the same bar graph:
In which two years was an equal number of wheat production observed?

(1) 2022 and 2023

(2) 2021 and 2023

(3) 2020 and 2021

(4) 2021 and 2022

✓ ANSWER: (2)

From the bar graph, wheat production was approximately equal in **2021 and 2023**.

Q.45

STATS

The number of leaves present in the following stem-leaf plot:

STEM	LEAF
1	2, 4, 5, 7
2	0, 1, 3
3	5, 9
4	3, 4, 5
5	2, 6, 7, 8, 9

(1) 5

(2) 4

(3) 12

(4) 17

✓ ANSWER: (4)

```
Stem 1: 4 leaves | Stem 2: 3 leaves | Stem 3: 2 leaves
Stem 4: 3 leaves | Stem 5: 5 leaves
Total = 4 + 3 + 2 + 3 + 5 = 17
```

Q.46

STATS

The formula to compute the mean for a set of n values $x_1, x_2, \ldots\ldots x_n$ is

(1) Mean $= \bar{x} = \frac{\Sigma x_i}{n}$

(2) Mean $= \bar{x} = \frac{\Sigma x_i}{n + 1}$

(3) Mean $= \bar{x} = \left(\frac{1}{x}\right)(\Sigma x - n)$

(4) Mean $= \bar{x} = \frac{1(\Sigma x + n)}{x}$

✓ ANSWER: (1)

The arithmetic mean is calculated as $\bar{x} = \frac{\Sigma x_i}{n}$ — the sum of all values divided by the total number of observations.

Q.47

STATS

The sum of deviations from mean ($\bar{x}$) is always __________

(1) 1

(2) 0

(3) 2

(4) −1

✓ ANSWER: (2)

A fundamental property of the arithmetic mean: $\Sigma(x_i - \bar{x}) = 0$. The sum of deviations from the mean is always zero.

Q.48

STATS

For the given set of data which of the following statements are correct?
Statement A: Mean is uniquely defined for a set of data
Statement B: Median does not depend on all the observations
Statement C: Mode is affected by extreme values

(1) A and B

(2) A and C

(3) B and C

(4) A, B and C

✓ ANSWER: (1)

Statement A is correct (mean is unique). Statement B is correct (median depends only on middle values). Statement C is incorrect — **Mode is NOT affected** by extreme values; the mean is.

Q.49

STATS

The formula to compute quartile deviation of a data set is __________

(1) $QD = \frac{Q_2 - Q_1}{2}$

(2) $QD = \frac{Q_1 - Q_3}{2}$

(3) $QD = \frac{Q_2 + Q_1}{2}$

(4) $QD = \frac{Q_3 - Q_1}{2}$

✓ ANSWER: (4)

Quartile Deviation (Semi-Interquartile Range) = $QD = \frac{Q_3 - Q_1}{2}$, where Q_3 is the third quartile and Q_1 is the first quartile.

Q.50

STATS

Match the following and choose the correct option.

LIST – I	LIST – II
(a) Range	(i) Middle number in sorted list of data
(b) Mean	(ii) Most often repeated value in a data set
(c) Mode	(iii) Difference between largest and smallest value of data set
(d) Median	(iv) Average of n observations

(1) a – iv, b – ii, c – i, d – iii

(2) a – iii, b – iv, c – i, d – ii

(3) a – iii, b – iv, c – ii, d – i

(4) a – i, b – iii, c – iv, d – ii

✓ ANSWER: (3)

Range (a) → (iii) Difference between largest and smallest; Mean (b) → (iv) Average; Mode (c) → (ii) Most repeated value; Median (d) → (i) Middle number in sorted list.

Q.51

STATS

Which of the following is not a Python package or library for science and numerical computations?

(1) Numpy

(2) Loops

(3) Matplotlib

(4) Pandas

✓ ANSWER: (2)

Loops is a programming construct (for, while), not a Python package. NumPy, Matplotlib, and Pandas are all popular Python libraries for scientific computing.

Q.52

STATS

Why is indentation important in Python?

(1) For Decoration

(2) To define blocks of code

(3) Increase speed

(4) To define libraries

✓ ANSWER: (2)

In Python, **indentation defines blocks of code** (like loops, functions, conditionals). Unlike other languages that use braces {}, Python uses whitespace indentation as syntax.

Q.53

STATS

In Python programming language, when a loop is used inside another loop, it is called as

(1) For loop

(2) While loop

(3) Nested for loop

(4) Inside loop

✓ ANSWER: (3)

A loop placed inside another loop is called a **Nested loop**. It can be a nested for loop or nested while loop.

Q.54

STATS

Write the output of this programme:

```
a = 5
b = 8
n = a**3 + b
m = n % 2
print (m)
```

(1) 0

(2) −1

(3) 1

(4) 2

✓ ANSWER: (3)

a = 5, b = 8

n = 5^3 + 8 = 125 + 8 = 133

m = 133 % 2 = 1 (remainder when 133 ÷ 2)

Output: 1

Q.55

STATS

Match List – I with List – II and select the correct answer:

LIST – I	LIST – II
(a) a = 20	(i) Complex number
(b) b = "Hello Python"	(ii) Integer
(c) c = 12.5	(iii) String
(d) d = 3.0 + 4.1j	(iv) Float

(1) a – ii, b – iii, c – i, d – iv

(2) a – iii, b – iv, c – ii, d – i

(3) a – ii, b – iii, c – iv, d – i

(4) a – i, b – ii, c – iii, d – iv

✓ ANSWER: (3)

a = 20 → (ii) Integer; b = "Hello Python" → (iii) String; c = 12.5 → (iv) Float; d = 3.0 + 4.1j → (i) Complex number.

Q.56

STATS

What is the output of the given programme?

```
x = 0
if x == 1:
    print ("A")
else:
    print ("B")
```

(1) B

(2) A

(3) error

(4) No output

✓ ANSWER: (1)

```
x = 0, and 0 ≠ 1
The condition (x == 1) is False
So the else block executes → Output: B
```

Q.57

STATS

Number of students in a class room is an example of __________

(1) Qualitative data

(2) Quantitative data and discrete data

(3) Ordinal data

(4) Quantitative and continuous data

✓ ANSWER: (2)

Number of students is **quantitative** (numerical) and **discrete** (whole numbers — you can't have 25.5 students).

Q.58

STATS

The fastest data collection tool is __________

(1) Interview

(2) Survey

(3) Google form survey

(4) Group discussion

✓ ANSWER: (3)

Google Form survey is the fastest data collection tool as it enables instant distribution to large groups, automatic response collection, and real-time data compilation.

Q.59

STATS

Match List – I (Examples) with List – II (Data types) and select the correct answer:

LIST – I (EXAMPLES)	LIST – II (DATA TYPES)
(a) Height of a person	(i) Nominal data
(b) Number of teachers in a school	(ii) Continuous data
(c) Aadhar Number	(iii) Ordinal data
(d) Student grades	(iv) Discrete data

(1) a – ii, b – i, c – iii, d – iv

(2) a – iv, b – ii, c – i, d – iii

(3) a – ii, b – iv, c – i, d – iii

(4) a – i, b – iii, c – iv, d – ii

✓ ANSWER: (3)

Height → (ii) Continuous; Number of teachers → (iv) Discrete; Aadhar Number → (i) Nominal (just an ID, no order); Student grades → (iii) Ordinal (A > B > C).

Q.60

STATS

The process of grouping the data according to their characteristics is called __________

(1) Organisation of data

(2) Classification of data

(3) Tabulation of data

(4) Presentation of data

✓ ANSWER: (2)

Classification is the process of arranging data into groups or classes based on their common characteristics or attributes.

04

CHAPTER 4

Information Technology

20 Questions · 20 Marks

DCET 2026 — Set D1

Q.61

IT

Find the output of the following program:

```
main ()
{
    int a = 125;
    Printf ("%d\n", 10 + a ++);
}
```

(1) 136

(2) 135

(3) 126

(4) 125

✓ ANSWER: (2)

```
a = 125
a++ is post-increment: uses current value first, then increments
10 + a++ → 10 + 125 = 135 (a becomes 126 after)
Output: 135
```

Q.62

IT

Which of the following is an example for valid identifier?

(1) $abcd

(2) 12_34a

(3) /M_ab

(4) abcd

✓ ANSWER: (4)

In C, valid identifiers must start with a letter or underscore. `abcd` is valid. $abcd ($ not allowed), 12_34a (starts with digit), /M_ab (/ not allowed) are all invalid.

Q.63

IT

Match standard symbols (List – I) used in program flowcharts with the purpose of use (List – II).

LIST – I (SYMBOLS)	LIST – II (PURPOSE)
(a) Rectangle	(i) Input/Output
(b) Diamond	(ii) Predefined processing
(c) Rectangle with lines	(iii) Computer Processing
(d) Parallelogram	(iv) Decision

(1) a – i, b – iii, c – iv, d – ii

(2) a – iii, b – iv, c – ii, d – i

(3) a – iii, b – ii, c – iv, d – i

(4) a – i, b – iv, c – iii, d – ii

✓ ANSWER: (2)

Rectangle → (iii) Processing; Diamond → (iv) Decision; Rectangle with double lines → (ii) Predefined processing/Subroutine; Parallelogram → (i) Input/Output.

Q.64

IT

Which of the following is not a valid component of an Android application?

(1) Activity

(2) Service

(3) Application

(4) Communication

✓ ANSWER: (4)

The four main Android components are Activity, Service, Broadcast Receiver, and Content Provider. **Communication** is not a valid Android component.

Q.65

IT

Match the following CSS property to the effect it has on the webpage.

LIST – I (CSS PROPERTY)	LIST – II (EFFECT)
(a) Padding	(i) Specifies the transparency level of an element
(b) Margin	(ii) Creates space inside an element, around the content
(c) Z-index	(iii) Creates space outside an element's border
(d) Opacity	(iv) Controls the vertical stacking order of elements

(1) a – iii, b – ii, c – iv, d – i

(2) a – ii, b – iii, c – iv, d – i

(3) a – ii, b – iii, c – i, d – iv

(4) a – i, b – iv, c – iii, d – ii

✓ ANSWER: (2)

Padding → (ii) space inside element; Margin → (iii) space outside border; Z-index → (iv) stacking order; Opacity → (i) transparency level.

Q.66

IT

Which of the following software applications is primarily used to read and render HTML web pages?

(1) Server

(2) Operating system

(3) Web browser

(4) Database

✓ ANSWER: (3)

A **Web browser** (Chrome, Firefox, Safari) is the software application specifically designed to read, interpret, and render HTML web pages.

Q.67

IT

Give the JavaScript method that removes the last element of an array

(1) Shift ()

(2) POP ()

(3) Slice ()

(4) Splice ()

✓ ANSWER: (2)

pop() removes and returns the last element of an array. shift() removes the first element, slice() returns a section, and splice() adds/removes at any position.

Q.68

IT

In a standard HTML form submission, what is the correct order of operations?

(a) Form validation (client-side)

(b) HTTP Request sent

(c) User clicks submit

(d) Server Response/Redirect

(1) a → c → b → d

(2) c → a → b → d

(3) c → b → a → d

(4) a → b → c → d

✓ ANSWER: (2)

The correct order: (c) User clicks submit → (a) Client-side form validation → (b) HTTP Request sent to server → (d) Server processes and sends Response/Redirect.

Q.69

IT

Match the functional areas of operation (List – I) with business function (List – II) with respect to Enterprise Resource Planning:

LIST – I (FUNCTIONAL AREA)	LIST – II (BUSINESS FUNCTION)
(a) Marketing and sales	(i) Purchasing goods and new materials
(b) Accounting and Finance	(ii) Recruiting and hiring
(c) Human Resources	(iii) Marketing a product
(d) Supply chain management	(iv) Financial accounting of payments from customers and to suppliers

(1) a – i, b – ii, c – iv, d – iii

(2) a – iii, b – iv, c – ii, d – i

(3) a – iii, b – ii, c – iv, d – i

(4) a – ii, b – iii, c – i, d – iv

✓ ANSWER: (2)

Marketing & sales → (iii) Marketing; Accounting & Finance → (iv) Financial accounting; HR → (ii) Recruiting/hiring; Supply chain → (i) Purchasing goods.

Q.70

IT

Which of the following modules of Enterprise Resource Planning handle payroll and employee related data?

(1) Supply chain Management

(2) Marketing Automation

(3) Human Resource Management

(4) Accounting and Finance

✓ ANSWER: (3)

The **Human Resource Management (HRM)** module handles payroll, employee records, attendance, benefits, and all employee-related data in an ERP system.

Q.71

IT

Choose the correct sequence of events for any sale with respect to SAP Enterprise resource planning sales.
(i) Pre-sales activities (ii) Inventory sourcing (iii) Billing (iv) Payment (v) Sales order processing (vi) Delivery

(1) (i), (iii), (iv), (ii), (v), (vi)

(2) (i), (iv), (iii), (v), (vi), (ii)

(3) (i), (v), (ii), (vi), (iii), (iv)

(4) (ii), (v), (iii), (vi), (iv), (i)

✓ ANSWER: (3)

SAP sales cycle: (i) Pre-sales → (v) Sales order processing → (ii) Inventory sourcing → (vi) Delivery → (iii) Billing → (iv) Payment.

Q.72

IT

Which of the following is a part of business process automation technology stack?

(1) Low-code development

(2) Android development

(3) Software design and development

(4) Hardware development

✓ ANSWER: (3)

Software design and development is a core component of business process automation technology stack, enabling creation of automated workflows and business solutions.

Q.73

IT

Which is the example of SaaS cloud service models?

(1) AWS Lambda

(2) Amazon EC2

(3) Google App Engine

(4) Gmail

✓ ANSWER: (4)

Gmail is a SaaS (Software as a Service) product — users access it via browser without managing infrastructure. AWS Lambda = FaaS, EC2 = IaaS, App Engine = PaaS.

Q.74

IT

Identify which of the following is NOT a characteristic of a public cloud?

(1) Multi-tenancy

(2) High availability and Redundancy

(3) Managed by a third-party provider

(4) Dedicated hardware for single user

✓ ANSWER: (4)

Dedicated hardware for single user is a characteristic of a private cloud, not public cloud. Public clouds share resources among multiple tenants.

Q.75

IT

Match List – I (Types of Cloud services) with List – II (service) and select the correct answer:

LIST – I (CLOUD SERVICE TYPE)	LIST – II (DESCRIPTION)
(a) IaaS	(i) Delivers software application over the internet
(b) PaaS	(ii) Allows developers to build applications without managing servers
(c) SaaS	(iii) Provides fundamental computing, network and storage resources
(d) Serverless computing	(iv) Offers a hosted environment for developers to build, test and deploy applications

(1) a – iv, b – iii, c – ii, d – i

(2) a – iii, b – ii, c – i, d – iv

(3) a – iii, b – iv, c – i, d – ii

(4) a – i, b – iii, c – ii, d – iv

✓ ANSWER: (3)

IaaS → (iii) fundamental computing resources; PaaS → (iv) hosted environment to build/test/deploy; SaaS → (i) software over internet; Serverless → (ii) build apps without managing servers.

Q.76

IT

Match List – I (Cloud deployment models) with List – II (operations) and select the correct answer:

LIST – I (DEPLOYMENT MODEL)	LIST – II (OPERATION)
(a) Public cloud	(i) Designed for shared needs of multiple organizations
(b) Private cloud	(ii) A combination of public and private clouds
(c) Hybrid cloud	(iii) Resources are used by a single organization
(d) Community cloud	(iv) Resources are owned and operated by third-party provider

(1) a – i, b – ii, c – iii, d – iv

(2) a – iv, b – iii, c – ii, d – i

(3) a – iii, b – iv, c – i, d – ii

(4) a – ii, b – iii, c – iv, d – i

✓ ANSWER: (2)

Public cloud → (iv) third-party owned; Private cloud → (iii) single organization; Hybrid → (ii) combination; Community → (i) shared needs of multiple organizations.

Q.77

IT

What mechanism does a firewall use to decide whether to allow or block network traffic?

(1) The physical location of the device

(2) Pre-defined security rules and criteria

(3) Type of operating system used

(4) Source code of the application

✓ ANSWER: (2)

A firewall uses **pre-defined security rules and criteria** to filter network traffic, allowing or blocking packets based on IP addresses, ports, protocols, and other parameters.

Q.78

IT

Which of the following is true about spear phishing?

(1) Spear phishing does not involve malicious links

(2) Phishing is only used for stealing credit card information

(3) Spear phishing targets a specific individual or organization

(4) Spear phishing uses telephone calls instead of emails

✓ ANSWER: (3)

Spear phishing is a targeted attack aimed at a specific individual or organization, unlike general phishing which targets many people randomly.

Q.79

IT

Match List – I (Cyber threats) with List – II (attacks) and select the correct answer:

LIST – I (CYBER THREAT)	LIST – II (ATTACK DESCRIPTION)
(a) Phishing	(i) Threatening to leak data
(b) Ransomware	(ii) Overwhelming server to disrupt services
(c) DoS attack	(iii) Compromising vendors to infiltrate target organization
(d) Supply chain attack	(iv) Attackers use deceptive emails to reveal credentials

(1) a – i, b – iii, c – ii, d – iv

(2) a – iii, b – iv, c – i, d – ii

(3) a – iv, b – i, c – ii, d – iii

(4) a – ii, b – iv, c – iii, d – i

✓ **ANSWER: (3)**

Phishing → (iv) deceptive emails; Ransomware → (i) threatening to leak data; DoS → (ii) overwhelming server; Supply chain → (iii) compromising vendors.

Q.80

IT

Which type of malware is capable of self-replication and spreading by creating copies of itself through infected systems?

(1) Trojan Horses

(2) Logic Bombs

(3) Spyware

(4) Computer worms

✓ **ANSWER: (4)**

Computer worms are malware that self-replicate and spread across networks without requiring user interaction or a host program, unlike viruses.

05

CHAPTER 5

Electrical & Electronics Engineering

20 Questions · 20 Marks

DCET 2026 — Set D1

Q.81
EEE

Identify the logic gate in which the output is high, when the two inputs are different

(1) AND gate
(2) OR gate
(3) XOR gate
(4) NAND gate

✓ ANSWER: (3)

The XOR (Exclusive OR) gate outputs HIGH (1) only when the two inputs are different. When both inputs are same (0,0 or 1,1), output is LOW (0).

Q.82
EEE

Choose the correct answers.
Sensors are used to perform the following functions.
(a) Control the actions as well as drive the plant directly
(b) To measure input signals for feed-forward control
(c) To measure process variables for system monitoring, diagnosis and supervisory control
(d) To generate control signals

(1) (a) and (b)
(2) (b) and (c)
(3) (c) and (d)
(4) (d) and (a)

✓ ANSWER: (2)

Sensors are primarily used to measure input signals for feed-forward control (b) and to measure process variables for system monitoring, diagnosis and supervisory control (c). They do not directly control/drive plants or generate control signals.

Q.83
EEE

PLC is an acronym for ___________.

(1) Pre-Logic Controller
(2) Programmable Logic Controller
(3) Programmable Logic Communication
(4) Post-Logic Controller

✓ ANSWER: (2)

PLC stands for **Programmable Logic Controller** — a digital computer used for automation of industrial processes.

Q.84

EEE

Identify the device used for the following conversion:
[i/p signal: AC sine wave → Device → o/p signal: DC pulsating wave]

(1) Inverter

(2) Rectifier

(3) Cycloconverter

(4) Chopper

✓ ANSWER: (2)

A **Rectifier** converts AC (alternating current) to DC (direct current). An inverter does the opposite (DC to AC).

Q.85

EEE

The block diagram with CPU, RAM, ROM, I/O, Timer, and Serial communication port represents the block diagram for

(1) Microprocessor

(2) Computer

(3) PLC

(4) Micro controller

✓ ANSWER: (4)

A **Microcontroller** integrates CPU, RAM, ROM, I/O ports, Timer, and Serial communication port on a single chip, unlike a microprocessor which requires external components.

Q.86

EEE

The material not used in fire extinguishers is ___________

(1) CO_2

(2) Foam

(3) Dry powder

(4) Nitrogen

✓ ANSWER: (4)

Common fire extinguisher materials include CO_2, foam, dry powder, and water. **Nitrogen** is an inert gas not used as a fire extinguishing agent.

Q.87

EEE

Pipe material used in earthing is __________

(1) Nickel & Aluminium

(2) Aluminium & Tungsten

(3) Tungsten & Galvanised Iron

(4) Galvanised Iron

✓ ANSWER: (4)

Galvanised Iron (GI) pipes are used for earthing because they resist corrosion, provide good conductivity, and are durable when buried in soil.

Q.88

EEE

Which one of the following is the conventional source of Electrical Energy?

(1) Solar Power

(2) Wind Power

(3) Tidal Power

(4) Thermal Power

✓ ANSWER: (4)

Thermal Power (coal/gas/oil-based) is a conventional energy source. Solar, wind, and tidal are non-conventional (renewable) energy sources.

Q.89

EEE

Which of the following electrical quantities and their SI units are correctly paired?
(a) Current — Ampere
(b) Emf — Volts
(c) Electrical Energy — Watts
(d) Resistance — Mho

(1) a, b, c and d

(2) Only a and b

(3) Only c and d

(4) Only b and c

✓ ANSWER: (2)

Current → Ampere ✓ and Emf → Volts ✓ are correct. Electrical Energy is measured in Joules (not Watts — that's Power). Resistance is measured in Ohms (not Mho — that's Conductance).

Q.90

EEE

Match List – I (Electrical Quantity) with List – II (Measuring Instruments):

LIST – I (ELECTRICAL QUANTITY)	LIST – II (MEASURING INSTRUMENT)
(a) Current	(i) Voltmeter
(b) Potential difference	(ii) Ammeter
(c) Power	(iii) Ohmmeter
(d) Resistance	(iv) Watt meter

(1) a – ii, b – i, c – iv, d – iii

(2) a – iv, b – i, c – ii, d – iii

(3) a – iii, b – iv, c – i, d – ii

(4) a – ii, b – iii, c – iv, d – i

✓ ANSWER: (1)

Current → (ii) Ammeter; Potential difference → (i) Voltmeter; Power → (iv) Watt meter; Resistance → (iii) Ohmmeter.

Q.91

EEE

Find the value of current 'I' in the circuit given:
[Circuit: Two 4Ω resistors in parallel, in series with 2Ω, powered by 4V battery]

(1) 2A

(2) 1A

(3) 1.5A

(4) 2.5A

✓ ANSWER: (2)

$$4\Omega \parallel 4\Omega = \frac{4\times4}{4+4} = 2\Omega$$

$$\text{Total } R = 2\Omega + 2\Omega = 4\Omega$$

$$I = V/R = \frac{4}{4} = 1A$$

Q.92

EEE

For the circuit given, calculate the voltage drop across 3Ω Resistor:
[Circuit: 3Ω in series with (6Ω ∥ 6Ω), powered by 6V]

(1) 6 V

(2) 3 V

(3) 1 V

(4) 2 V

✓ ANSWER: (2)

$$6\Omega \parallel 6\Omega = \frac{6\times6}{6+6} = 3\Omega$$

Total R = 3Ω + 3Ω = 6Ω

$$I = V/R = \frac{6}{6} = 1A$$

Voltage across 3Ω = I × R = 1 × 3 = 3V

Q.93

EEE

The function/s of fuse wire is/are to:
(a) Carry normal working current without heating
(b) Break the circuit during short circuit
(c) Protect the circuit from undervoltage
(d) Protect the circuit from overvoltage

(1) a, b & c

(2) b only

(3) a & b

(4) a, b & d

✓ ANSWER: (3)

A fuse wire (a) carries normal current without heating and (b) breaks the circuit during short circuit/overcurrent. It does NOT protect against undervoltage or overvoltage.

Q.94

EEE

Rating of a fuse wire is ___________

(1) Maximum current at which a fuse burns

(2) Maximum voltage at which the fuse burns

(3) Maximum current which fuse can carry without burning

(4) Minimum current at which fuse burns

✓ ANSWER: (3)

Fuse rating = **maximum current the fuse can safely carry without blowing/burning**. If current exceeds this rating, the fuse melts and breaks the circuit.

Q.95

EEE

Match the following related to fuses and their applications.

LIST – I (FUSE TYPE)	LIST – II (APPLICATION)
(a) Glass cartridge fuse	(i) Emergency protection of electrical circuits
(b) Kit Kat Fuse	(ii) Power distribution systems
(c) HRC Fuse	(iii) Electronic equipments
(d) Fuse wire	(iv) Residential, small scale industrial applications

(1) a – ii, b – iii, c – iv, d – i

(2) a – iii, b – iv, c – ii, d – i

(3) a – iv, b – i, c – ii, d – iii

(4) a – iii, b – ii, c – iv, d – i

✓ **ANSWER: (2)**

Glass cartridge fuse → (iii) Electronic equipment; Kit Kat Fuse → (iv) Residential/small industrial; HRC Fuse → (ii) Power distribution; Fuse wire → (i) Emergency protection.

Q.96

EEE

If N_1 represents primary windings and N_2 represents secondary windings of transformer, match:

CONDITION	TRANSFORMER TYPE
(a) $N_1 = N_2$	(i) Auto transformer
(b) $N_1 < N_2$	(ii) Isolation transformer
(c) $N_1 > N_2$	(iii) Step-up transformer
(d) Only one winding	(iv) Step-down transformer

(1) a – ii, b – iii, c – iv, d – i

(2) a – i, b – ii, c – iii, d – iv

(3) a – iii, b – ii, c – iv, d – i

(4) a – ii, b – i, c – iii, d – iv

✓ **ANSWER: (1)**

$N_1 = N_2$ → (ii) Isolation transformer (1:1 ratio); $N_1 < N_2$ → (iii) Step-up (increases voltage); $N_1 > N_2$ → (iv) Step-down (decreases voltage); One winding → (i) Auto transformer.

Q.97

EEE

Identify the properties of ideal transformer.
An ideal transformer is one that has:
(a) No winding resistance
(b) No leakage flux
(c) No iron losses
(d) High winding resistance

(1) (a), (b) and (c)

(2) (b), (c) and (d)

(3) (c) and (d)

(4) Only (d)

✓ ANSWER: (1)

An ideal transformer has: (a) zero winding resistance, (b) no leakage flux, and (c) no iron losses — meaning 100% efficiency. (d) High winding resistance is NOT a property of an ideal transformer.

Q.98

EEE

A 1000/100V, 20 kVA transformer has 50 turns in the secondary. Calculate the primary turns.

(1) 50

(2) 500

(3) 5

(4) 5000

✓ ANSWER: (2)

$$\frac{V_1}{V_2} = \frac{N_1}{N_2}$$

$$\frac{1000}{100} = \frac{N_1}{50}$$

$$N_1 = 50 \times 10 = 500 \text{ turns}$$

Q.99

EEE

The working principle of transformer is based on ___________.

(1) Coulomb's law

(2) Faraday's Law

(3) Newton's Law

(4) Ohm's Law

✓ ANSWER: (2)

A transformer works on **Faraday's Law of Electromagnetic Induction** — a changing magnetic flux through a coil induces an EMF in the coil.

Q.100

EEE

The capacity of battery is primarily measured in __________

(1) Voltage (V)

(2) Power (W)

(3) Ampere-hours (AH)

(4) Current (I)

ANSWER: (3)

Battery capacity is measured in **Ampere-hours (Ah)** — it indicates how much current a battery can deliver over a specific time period.

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